Safety and Accessibility study at the tourist places in Delhi

Major Project Dissertation

Submitted by

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DECLARATION

This is to certify that the work that forms the basis of this project Safety and Accessibility study to the tourist places in Delhi is an original work carries out by me and has not been submitted anywhere else for the award of any degree.

I certify that all sources of information and data are fully acknowledged in the project dissertation.

Sattaru Manohar

Date: 01/06/2020

CERTIFICATE

This is to certify that Sattaru Manohar has carried out his major project in partial fulfillment of the requirement for the Degree of Master of Technology in Urban Development and Management on the topic Safety and Accessibility study to the tourist place in Delhi during January 2020 to May 2020. The Project was Carried out at the SAFETIPIN Organization- Gurugram.

This dissertation embodies the original work of the candidate to the best of our knowledge.

Date 27/05/2020

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List of Abbreviations

UNWTO - United Nations World Tourism Organization

UNESCO – United Nations Educational, Scientific and Cultural Organization

WHO - World Health Organization

MOSPI - Ministry of Statics and Program Implementation

GOI - Government of India

MOT - Ministry of Tourism

ASI - Archeological Survey of India

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ABSTRACT

Safety and accessibility at the tourist destinations gained importance globally due to increasing safety issues and accessibility challenges. World tourism organization come up with set of guidelines and indicators to improve the tourism industry. Developing countries like India are also facing challenges to provide accessible infrastructure at the tourist destinations. This study attempted to audit 10 tourist places in Delhi includes UNESCO identified historical sites, religious places and markets with a set of safety and accessibility parameters based on the literature review and guidelines by the Ministry of tourism, GOI. A set of questionnaires are prepared based on the parameters to audit the different locations like Outside, at the parking, at the ticket counter, at the gate entry and inside the tourist places in google forms and linked to Safetipin applications.

In this Study MySafetiPin and SafetiPin Site applications are used to audit the tourist place and the same has been done in the pilot survey to test the flow of data collection. Due to circumstances the entire study done on the basis of secondary data is collected through safetipin organization for safety parameters and tourist places are audited through virtual tours, Google map street views and youtube videos for accessibility parameters. Key findings are some tourist places like Qutub Minar, Red fort, Bangla sahib Gurudwara, Purana Qila, Lotus temple, Jantar mantar, Dilli Haat INA are equipped with accessible infrastructure and safe to visit. Tourist place like Tughlakabad fort is not safe to access due to lack of accessible infrastructure and lack of security surveillance inside. This study can be further used to develop a holistic approach to auditing of tourist places and the development of comprehensive plans and tool kits to assist planners and policy makers in setting benchmarks and measure compliance to the benchmarks.

Key words: Safety, accessibility, tourist attractions, Infrastructure auditing

Chapter 1

Introduction to Tourism

Tourism can be defined as a group of activities and services which deliver a travel experiences includes mobility, hospitality, entertainment and other related services provided to individual or a group of people travelling away from home. Basically, categorized as domestic, Inbound and outbound tourism which means any individual or a group of people who explore in their country considered as domestic tourism where as if the same group travels to other country as outbound and if any foreigners arrives to country considered as inbound tourism.

According to UNWTO tourism and travel industry emerged as world's largest and dynamic industry contributing 10% of the global GDP and even some countries like Maldives, Macao, Cambodia, Thailand economy majorly depends on Tourism industry. In the process of globalization quality of services improved decreased in travel time with good connectivity, accessibility to the tourist destination and information systems. Due to increase in household income of individual and purpose of travel like recreation, holydays, business and professional trips are interconnected with economic activities that transformed societies and improved quality of living which creates networks of various activities in different sectors accommodations and enabled service and food supply chains, helps in development of accessible infrastructure which again benefits the dependent communities.

Now a day's safety and security is the main concern in majority of tourist visitor due to natural disasters, terrorism, crime and harassment by tourists as well as local people at the tourism destinations and mostly the victims are women and children. This will impact the overall development of the region as well as dependent host communities. According the WHO 15% of the world population facing different disability challenges. Providing continuous accessible infrastructure services for the disabled tourist visitors is another challenge to the governments and local municipal bodies and the tourism departments. In some situations, due to lack of infrastructure and surveillance may trigger the safety and security issues can be solved by auditing the spaces and identifying the issues through research.

1.1 Tourism sector in Delhi - India

Tourism sector in India is growing rapidly and improving the country GDP with a contribution of 9.4 lakh crore industry. In 2019 India ranked 34th out of 140 countries and particularly scored 4.5 and 2.8 in safety security and tourist service infrastructure out of 7 in the TTCI report. Travel and Tourism Competitiveness Index report(TTCI) is developed on four pillars and includes sub parameters in which safety and security is a sub parameter of enabling environment and tourist service infrastructure as a part under infrastructure service in the overall South Asia region India is attracting the maximum tourist population destination from the global tourism for health and cultural tourism.

According to ministry of Tourism report in India Delhi have a highest foreign tourist arrival with a share of 25.46% and Currently Delhi NCR is 5th world largest urban agglomeration and in 2020 holding a population is now estimated at 30,290,936 and infrastructure like International airport, better connectivity to World heritage sites and also to access other tourist places in Himachal Pradesh, Uttarakhand, and Kashmir and South India as well.

In the mean while the concerns regarding safety and security was increasing due to various incidents that happened in the past decade which may impact the foreign arrivals to the country.

"Atithi Devo Bhava" in our culture itself we pray guests as gods but in the process of urbanization and development we forgot our roots.

A think thanks of Government of India, Bureau of Police Research and Development proposed a Tourist policing in 2017 to support both international and national tourist at the tourist spots but still some physical factors which are dependent on space and time influencing the quality of experience and a study is required on indicators that may support to increase the safety and security to women around the tourist places in Delhi.

1.2 Introduction to Safety and Accessibility at tourist destinations

The main concern of any tourist visitor is safety and security and important in any country across tourism linked sectors and it may affect the foreign tourist arrivals. To improve the condition of safety and security at the tourist destinations government introduced the tourist policing at the destination in 2010 which guides the tourists by facilitating in finding safe accommodations, fast handling enquires and complaints, quick actions on the security issues and continuous surveillance at the tourist places. Ministry of tourism also focused on tourist visitor health by enhancing the sanitation facilities and clean drinking water even training the staff for life support at emergency situations. Improving the road and transport infrastructure to reduce the accidents on roads and decrease the travel time. Archeological survey of India looks after the historical tourist sites in India. Even focuses on reducing the harassment, crime and violence against women enforcing women police and women security and staff at the tourist destinations (Subash, 2015).

In growing economy like India providing accessible infrastructure is another challenge at the tourist destinations. To make tourism accessible to all Ministry of tourism and GOI given guideline to state government to develop accessible infrastructure at the destinations for this they come with the accessibility guidelines to follow and work on universal design strategy like pre visit information, online ticket bookings and providing information about accessibility services. Capacity building to tourist staff and guides in handling disabled visitors. Government also introduced HOHO bus network connecting different tourist places in Delhi.

Installing ramps, handrails and tactile pavements for continuity and even suggested to keep wheel chairs and audio guides. Guidelines for installing signages and tourist site maps printed on tails. Accessible infrastructure at the ticket counters and at the gate entry. Access to maximum parts at heritage and historical sites emergency evacuation plans and multipurpose designs to access public infrastructure (MOT and GOI, n.d.).

Chapter 2

Literature review:

In this paper authors shared hypothetical bits of knowledge and managerial reactions identified with security issues and security emergency. In the last two decades after 9/11 terrorist attack concerns related to safety and security gained importance in global tourism administration and researchers started working on management tools to all partners in all levels. Due to development of global connectivity, increased income levels, travel, in addition nations of the supposed third world. for them the travel industry is a piece of their financial advancement procedure over a time period travel industry itself got probably the biggest business. Required a detailed research on various measurements in wellbeing and security like open security, health, sanitation, lawful insurance of vacationers in tourism industry (Kôvári, n.d.)

In this paper authors done an assessment after 3 years of implementation of the tourism indicators from 2013 to 2016 in the selected tourism regions and prepared a set of survey questions based on the destination management, social and cultural impact, economic value and environmental impact initially there was no statistical data available at the local level and in this process of collecting statistical data which takes more time. the recommends need inside future examinations for reinforcing the contribution of the private division in actualizing the STPMS. essentially, a nearby managers exhibited various degrees of enthusiasm for understanding the arrangement of markers, uncovering another territory for future research to make a common vision of maintainable the travel industry. this requires the formation of a system for guideline goal the executives and it is necessary to encourage the usage of standard marker frameworks to build sustainability arrangements and encourage arranging process. the marker frameworks should likewise empower the improvement of commonsense methodologies that help the basic utilization of pointers framework. the execution of best practices, through evaluation of marker frameworks, can be a huge help for both open and private procedures. (Patrizia Modica, 2018)

Author states that there are three distinct strategies to measure accessibility to Tourist spots Infrastructure, activity-based measures and utility opportunities measures. the segments of accessibility to tourist spots are four segments: land use, transportation, individual characteristics and transient components.

Subsequently to the provincial level or the entire nation level activity based measures have better than measure accessibility to the tourist places, especially the area based measures and event of the traveler action, in that goal makes spatial connection between the different visitor movement, in that goal, makes spatial relation between the different vacation spots through utilizing available vehicle modes. in this way gravity based and separation appear to be the best measures to evaluate accessibility of tourist destinations through the road networks by various vehicle modes. (Saad Jubran H. Al Kahtani, n.d.)

In this paper author made an attempt to establish a relationship between the direct impact of GDP and foreign tourist arrivals over a time period due to lack of Women's safety, poor infrastructure. In a study done by BBC major concern was sanitation and followed by safety and security. There are so many incidents happened against the international tourists especially foreign women and this spread as negative publicity for India. (Sayantani Basak, 2015)

Safetipin done safety auditing on 10 tourist places includes world heritage sites in Delhi through its application platform "My safetipin" and safety audits was done and analysis of infrastructure and a score is given on the basis of nine parameters lighting, Walkpath, Security, Public Transport, openness, visibility, people, , gender usage, feeling. (Safetipin, 2016)

In this paper author discussed the importance of safety and security factor that contributes to the tourist destination on basis of competitiveness index in Serbian region. Safety issues raised by the political instability, terrorists, harassment and organized crimes and natural disasters directly impacts the decrease in the tourist visitors and also local economic activities. Even unhygienic condition at the destination like drinking water, public toilets and food available also arises the health issues. according to TCCI safety and security indicators scored on basis of Business costs of violence and crime, Terrorism, Reliability of police service, Homicide rate and Index of terrorism incidence. In conclusion it is important to identify the safety and security risks and finding interventions can increase level of satisfaction creates a good opinion on the tourist destinations (Ninela Kordić*, 2015) .

In the paper author attempted to capture the experience of a tourists visited Brasov country tourism destinations and gathered quantitative opinion related to overall accessibility based on the survey and used multistage and simple random sampling method. Key finding like 27% of total sample finds difficult to access at the destination sites and 16% of sample feels lack of physical infrastructure (Gabriel B, 2016).

UNWTO published a manual consists of 5 modules and also given a overview regarding accessible tourism for all. This manual stated that due to increase in physically disabled population a customized frame work is required to improve the barrier free infrastructure by auditing, planning universal designs based on the requirements. Local governments and tourism management companies are still thinking that installing this infrastructure requires huge investments and lack of political will too. But by improving the environments can tap a potential in this accessible tourism sector and suggested to identify stakeholders and local entrepreneurs to provide the accessible services. In conclusion different tourist places requires customized solutions which can be done through including disabled persons to audit the tourist attraction and plan to focus on place making is added advantage both social and economical perspective.

Chapter 3

3.1 Aim and objectives and methodology

Aim: To assess the safety and accessibility at the tourist places in Delhi with respect to selected parameters.

Objectives:

1. To study safety and accessibility at the tourist places in Delhi

2. To develop a methodology to audit the tourist places based on safety and accessibility parameters.

3. To find the safety and accessibility gaps at the tourist places.

Research Questions:

1. What are factors that affect the safety and accessibility at the individual tourist place?

Methodology:

Background study: Study of various news articles, journals, publications, reports and other relevant resources.Training on coding image data to analyze the data collected through safetipin applications.

Selection of study area: Rationality behind choosing the particular tourist places depending upon their closing timings, type of tourist place and the data gathered from Delhi tourism website.

Literature review: Study of relevent reports, guidelines, journals, news articles and publications.

Data collection : Preparing questionnaires based on the safety and accessibility parameters, uploading in google forms and linking to safetipin site application. Preparing questionnaires based on the safety and accessibility parameters, uploading in google forms and linking to safetipin site application. Secondary data collection through MySafetipin application, virtual tours, google maps street view and tourist place youtube videos.

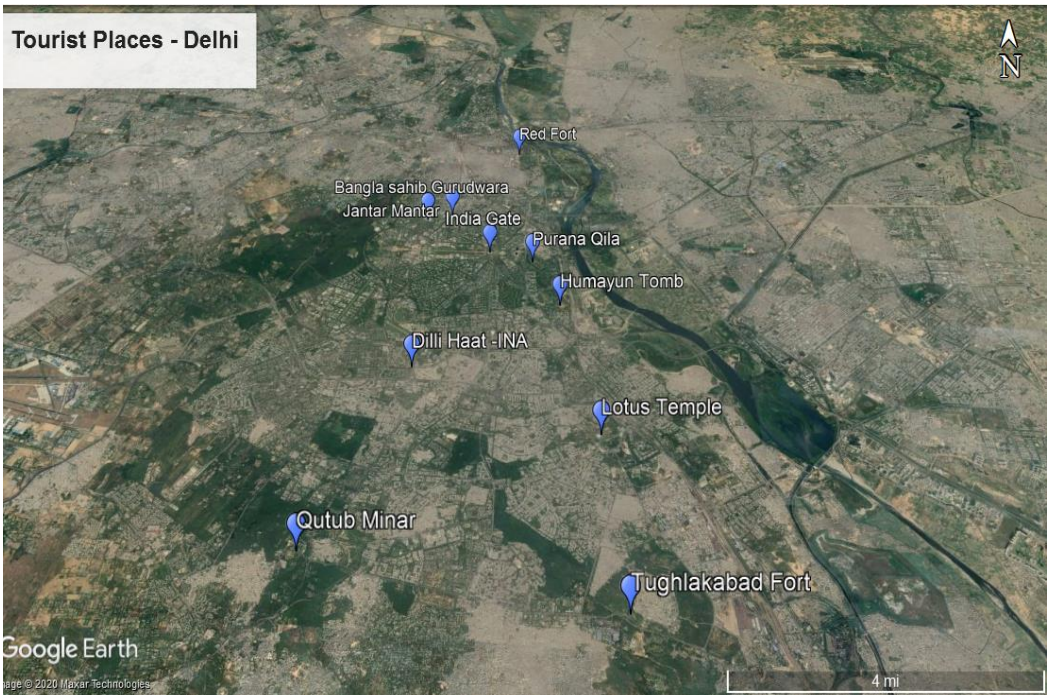
Data Analysis: Converting the data into excel sheets from the virtual audits and coding the captured images. Quantitative and qualitative analysis of the data with excel tools and Qgis for representing the audit locations in maps.

Findings and recommendations: Based on the findings recommendations are given to improve the safety and accessibility parameters at the selected tourist places.

Study area:

Delhi is the study region due to its significant history and house for world heritage sites and tourist recreational spaces. 10 tourist sites are selected on the basis of type of social usage, historical significance, entry fee, closing timings through Delhi tourism portal.

Map 1: Locations of the tourist places in Delhi



Source: Author

Table 1: Selected tourist places in Delhi

Name of Tourist place	Closing tim	Entry Fee	After 6 pm	Type
Lotus temple	*5:30*	*N*	*N*	Temple
Dilli Haat- INA	10:00	Y	Y	Market
Hymayun Tomb	6:00	Y	N	World heritage site
India Gate	0:00	N	Y	War Memorial
Jantar Mantar	6:00	Y	N	Historcal mounment
Puruna Quila	5:00	Y	Y	Fort
Qutib Minar	10:00	Y	Y	World heritage site
Bangla Sahib Gurudwara	0:00	N	Y	Temple
Red fort	4:30	Y	Y	World heritage site
Tuglakabadfort	5:00	Y	Y	Fort

Source: Author, 2020

3.2 Tools and Data collection methods

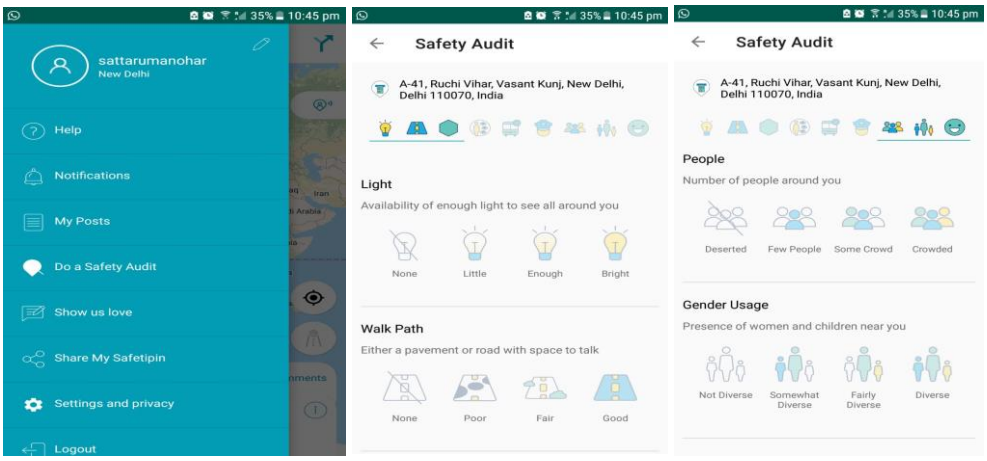
Tools:

Data collection - Safetipin site: Auditing the spaces can be done using Safetipin site in a systematic way which is interlinked with google forms.

Figure 1: Safetipin site application for auditing public spaces

Source: Safetipin

Figure 2: Safety audits should be done through MySafetipin application



Source: Safetipin

Google forms: Questions uploaded in google forms to take site survey outside and inside the tourist place.

Excel: To analyze the data which is collected through audits and virtual tours and performing calculations like weighted average etc. and to produce the graphs.

Data collection Method:

- Auditing the tourist places through virtual tours, Google street maps, Tourist place videos uploaded in Youtube.

- Capturing maximum possible data in image formats and coding these images based on the questionnaires.
- Converting the coded data into excel sheets and cross checking the sources.

Safety and Accessibility Parameters:

The Parameters are Identified on basis of literature review and guidelines suggested by the organization and national governments. So basically, safety parameters are considered from safetipin at three different locations.

Table 2: Safety parameter at the different locations

Safety	**Outside**	**At Gate**	**Inside**
Lighting	Applicable	Applicable	Applicable
Openness	Applicable	Not Applicable	Applicable
Visibility	Applicable	Applicable	Applicable
Crowd	Applicable	Applicable	Applicable
Security	Applicable	Applicable	Applicable
Walkpath	Applicable	Applicable	Applicable
Public Transport	Applicable	Not Applicable	Not Applicable
Gender Usage	Applicable	Not Applicable	Applicable
Feeling	Applicable	Not Applicable	Applicable

Source: Author

- Applicable indicates that particular parameter data is collected on different location based on the requirement.
- Not Applicable indicates parameter data is not necessary at that location.

Table 3: Accessibility audit at the different locations.

Accessibility	**Outside**	**At Gate**	**Inside**
Bus stop	Applicable	Not Applicable	Not Applicable
Parking	Applicable	Not Applicable	Not Applicable
Ticket counter	Applicable	Not Applicable	Not Applicable
Public toilets	Applicable	Not Applicable	Applicable
Drinking water	Applicable	Not Applicable	Applicable
At buildings and monuments	Not Applicable	Not Applicable	Applicable
Walkpath	Applicable	Applicable	Applicable

Source: Author

Auditing done at these places to check proper infrastructure like tactile pavements, handrails, ramps exists or not and also check the continuity to access the different locations.

Chapter 4

Results and discussions

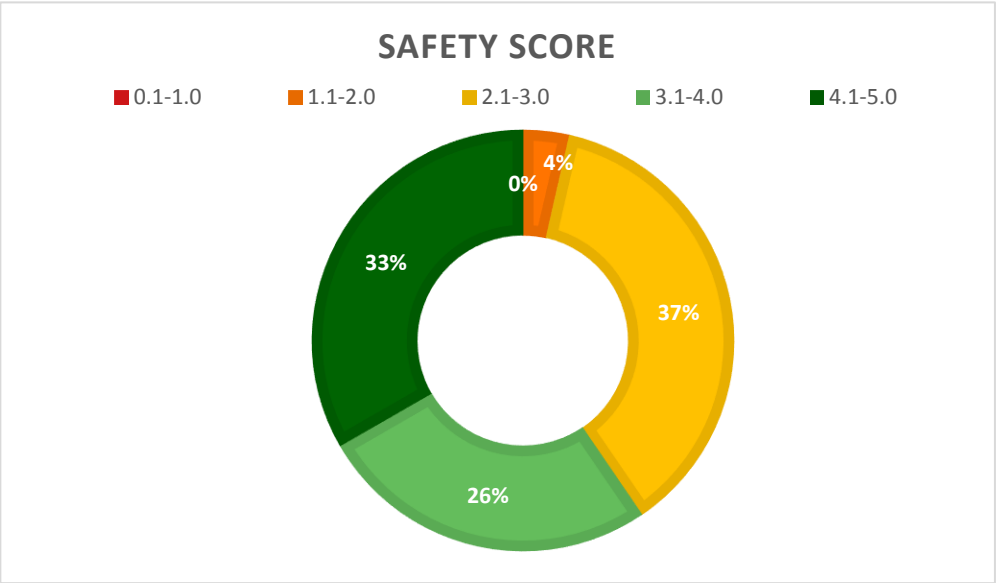
4.1 Qutub Minar

Qutub was built in 13th century and is located south of the Delhi. UNESCO identified this area as one of the world heritages sites in India. Minar was constructed with red sandstone and surrounding archaeological area contains burial sites. In current times qutub minar is one of the busiest tourist places in Delhi.

Safety Score - Outside the tourist place

Overall safety score was rated as 3.5 which is above average and total 84 T- audit points.

Figure 3: Safety Score - Qutub Minar



Source: Author

when it comes to accessibility, ticket counter is well accessible with tactile pavements and handrails and even tickets can be booked online to access the tourist place.

Private security is available at the entry of gate including women security, limited wheel chairs are available for old and physically challenged visitors and clock room available to keep the excess luggage for the tourists.

Most of the visitors visits in between from morning 10 am to evening 7 pm. Security also available in different parts of the tourist place.

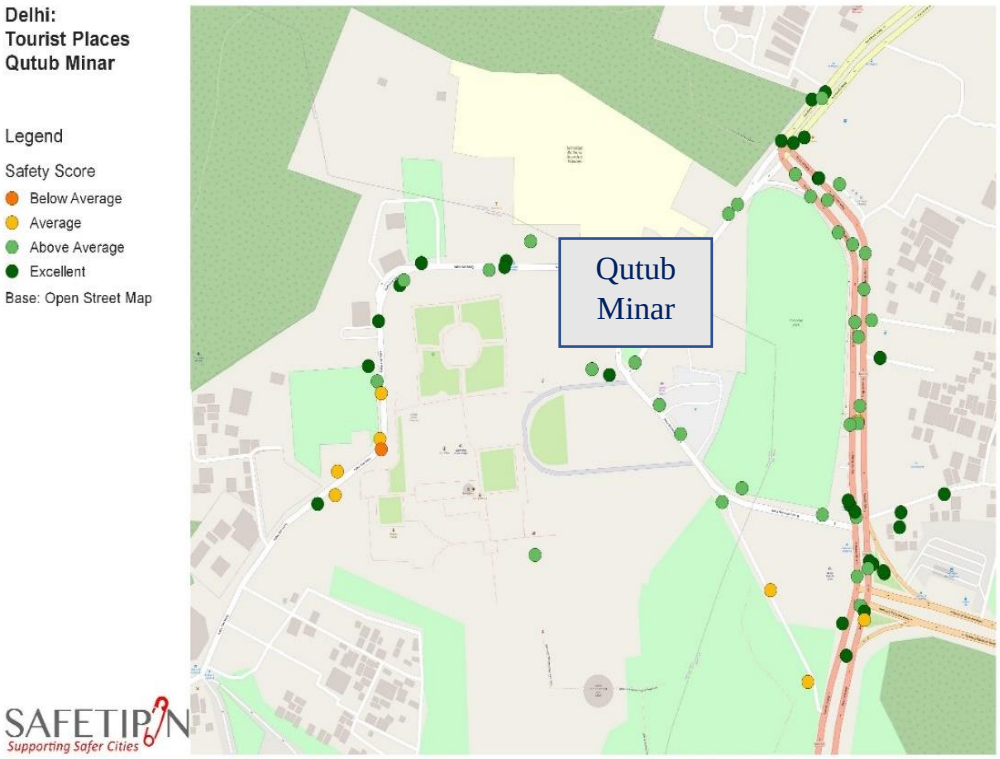
Figure 4: Ticket counter- paved walk path and handrails

Source: Author

Inside the tourist place most of the parts are accessible well equipped with ramps and tactile and handrails.

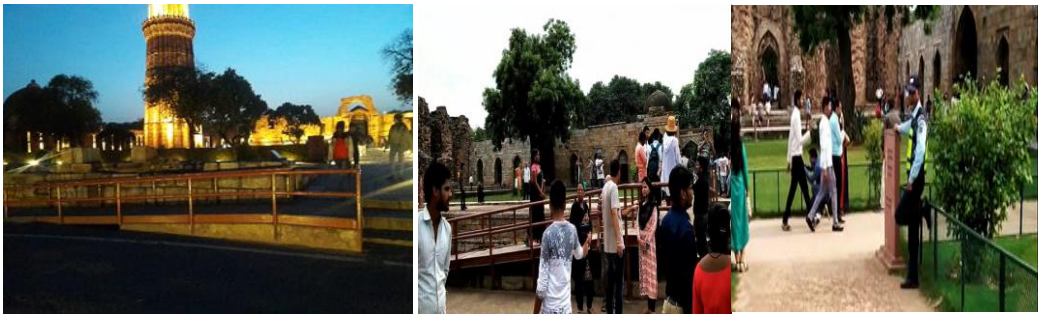
In the below map, points located around the tourist place as seen in the Safety Score map

Map 2: Indicating Safety Score ratings around Qutub Minar



Source: Safetipin

Figure 5: Ramps and handrails to access the different locations inside.



Source: Author

Signage boards are available to access the drinking water and public toilet facilities which are accessible.

Figure 6: Accessible infrastructure to drinking water and public toilet facilities.

Source: Author

Outside this tourist place have police post and patrolling by tourist police and dedicated parking places for the vehicles with surveillance, security check at the parking entry exits and also have accessible drinking water kiosk.

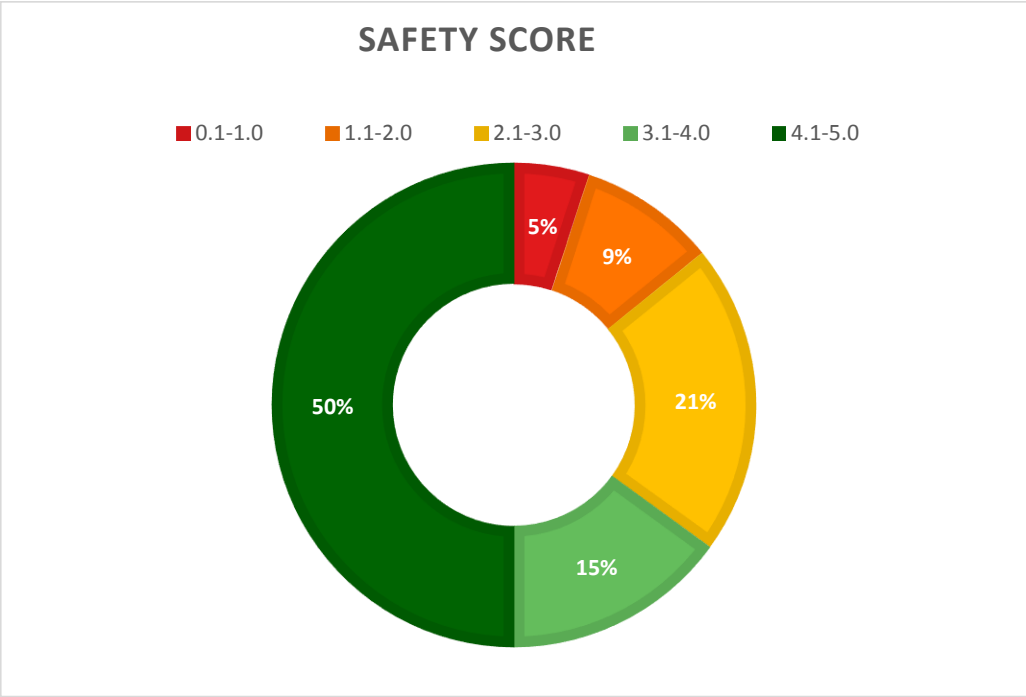
4.2 Dilli Haat – INA

Dilli Haat- INA is a market place set by the Delhi Tourism and Ministry of Tourism which provides traditional hand crafts, hand crafted fabrics and textile, cultural and regional foods which replicates rural market ambience. Aims to encourage artists from all over the country in order to support the rich heritage and culture of India known as unique craft bazar in the heart of National capital Delhi inaugurated in the year 1994. It is located near to Dilli Haat-INA metro station one of the tourist place and recreational public space.

Safety Score - Outside the tourist place

Overall safety score was rated as 3.5 which is above average and total 120 T- audit points.

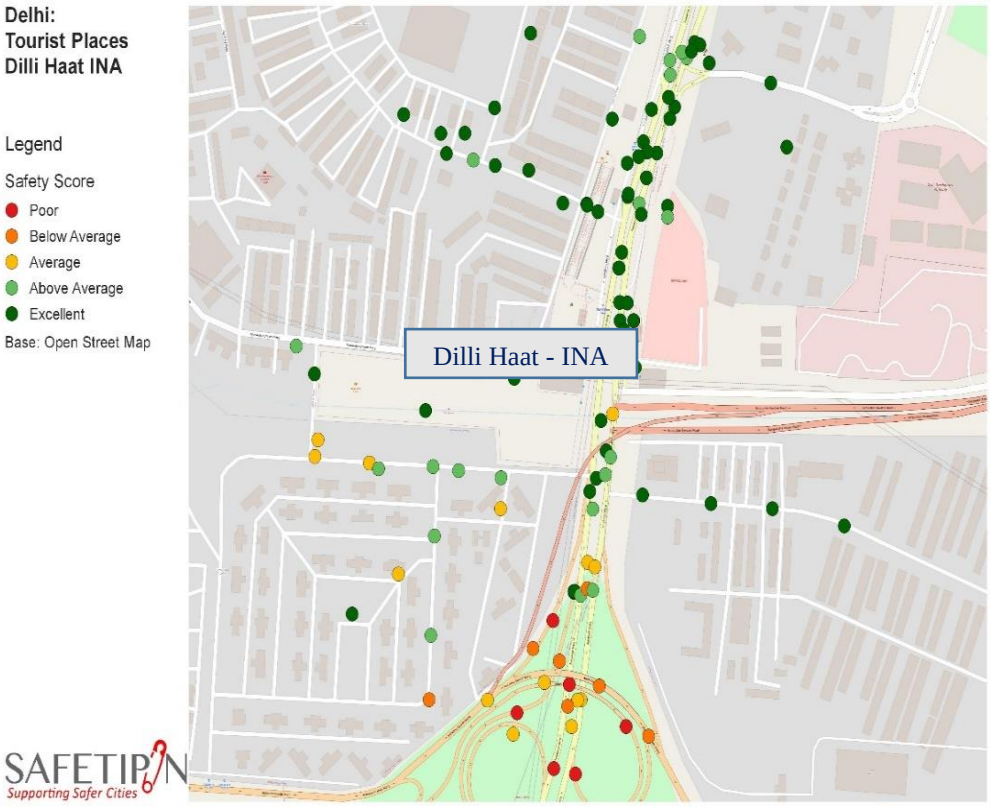
Figure 7: Safety Score – Dilli Haat INA



Source: Author

In the below map, points located around the tourist place as seen in the Safety-Score map

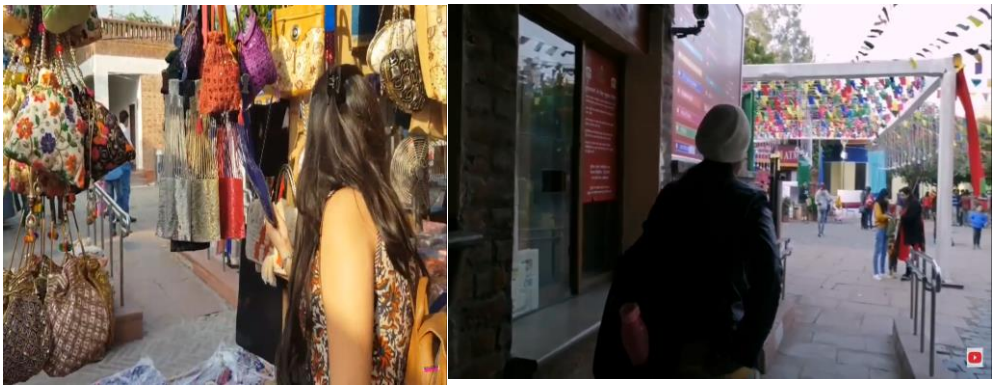
Map 3: Indicating Safety Score ratings around Dilli Haat INA



Source: Safetipin

Ramp and handrails at the ticket counter and inside the market place, security check at the entry gates and parking facilities available near the gate. Public toilets are accessible inside the market.

Figure 8: Ramps and handrails at the ticket counter and inside.



Source: Youtube, Virtual Audit

4.3 Lotus temple

Lotus temple is located in the East of Nehru place constructed in the shape of a lotus flower which is one of the seven major Baha's temples around the world completed in 1986 with a well-planned green landscaped garden even the structure is made with pure white marble. Tourist from any religion can visit and pray or meditate inside the temple. Entry is free and security check at the entry and the entire place is secured with private and police guards.

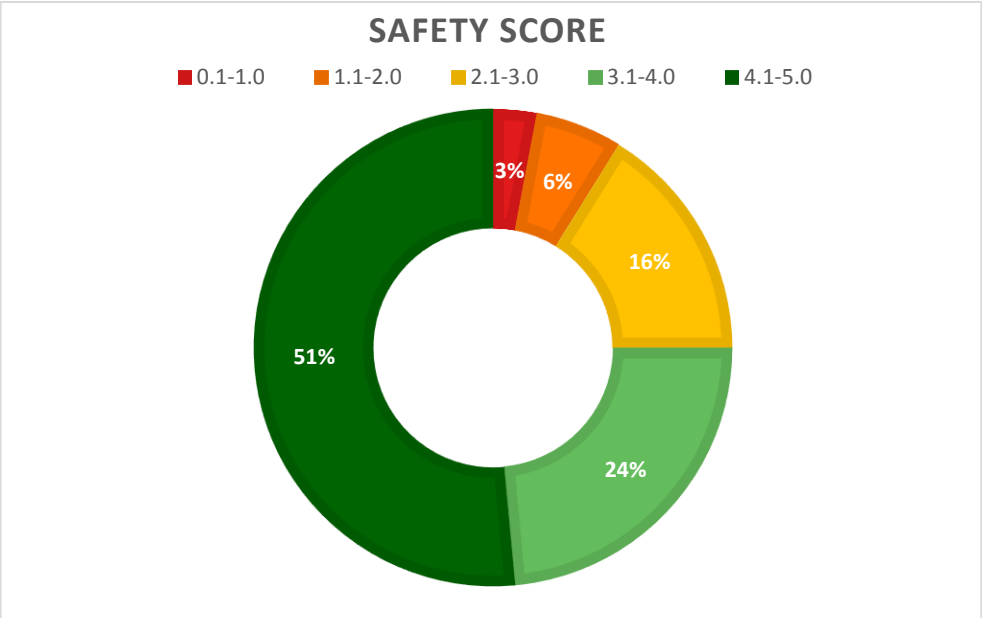
Figure 9: Security check and the entry and private security inside lotus temple.

Source: Youtube, virtual audit.

Safety Score - Outside the tourist place

Overall safety score was rated as 3.7 which is above average and total 69 T- audit points.

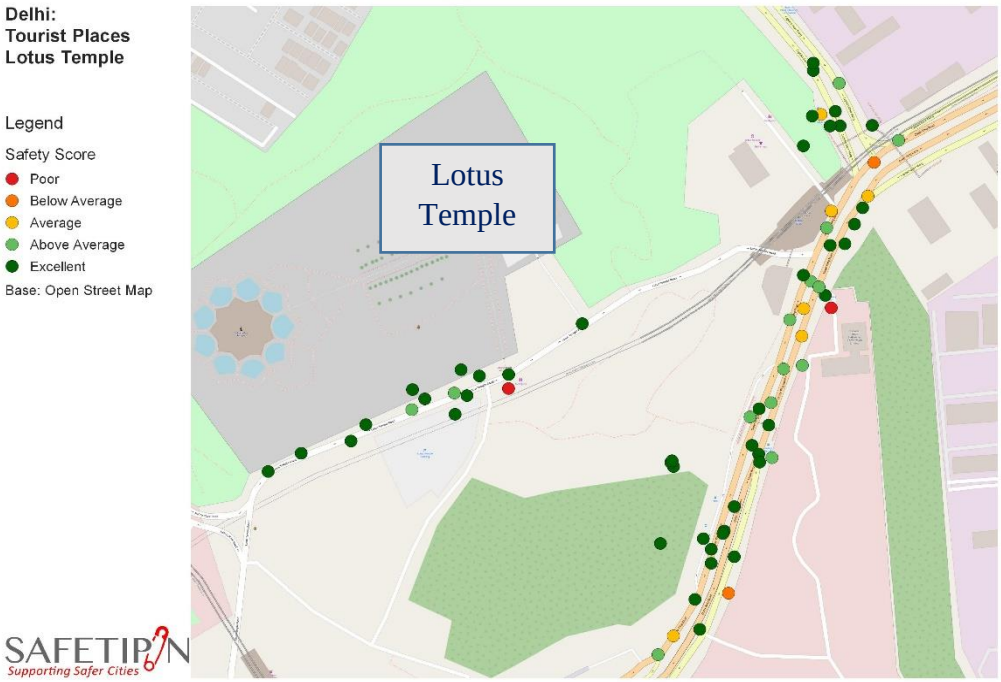
Figure 10: Safety Score – Lotus Temple



Source: Author

In the below map, points located around the tourist place as seen in the Safety Score map

Map 4: Indicating Safety Score ratings around Lotus Temple



Source: Safetipin

Dedicate parking facilities are available at this tourist place and tourist police and police post is located near the gate, Issues like street vendors encroached the walkpath and on the road too.

Figure 11: Presence of police post and walkpath encroached by street vendors.

Source: Youtube, Virtual Audit.

Ramps and handrails are available to enter inside the lotus temple and to the green landscapes. Overall, this tourist place is accessible and highly secured, but didn't noticed public toilet and drinking waters facilities inside.

4.4 Red Fort

The Red fort also known as Lal Qila was Constructed in Mid-17th Century by the 5th Mughal emperor Shah Jahan on the bank of the river Yamuna. This fort was built with Red sand stone enclosing with walls and replicates the creativity of Mughal for its planning and architectural design and landscapes. UNESCO announced this fort as world heritage site in 2007 and is managed by ASI. Now a days it is one of the highly secured tourist attractions in the heart of Delhi connected with metro and public bus transport. In the evening sound and light show events in India's history related to the red fort.

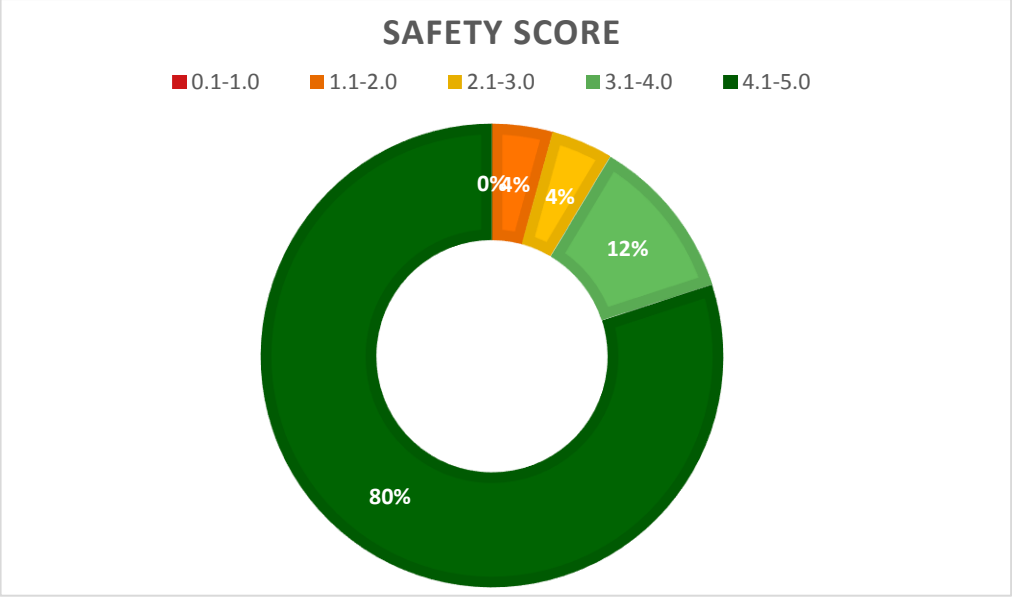
Figure 12: Hand rails at the ticket counter and security check at the entry

Source: Youtube, Virtual audit.

Safety Score - Outside the tourist place

Overall safety score was rated as 4.2 which is good and total 70 T- audit points.

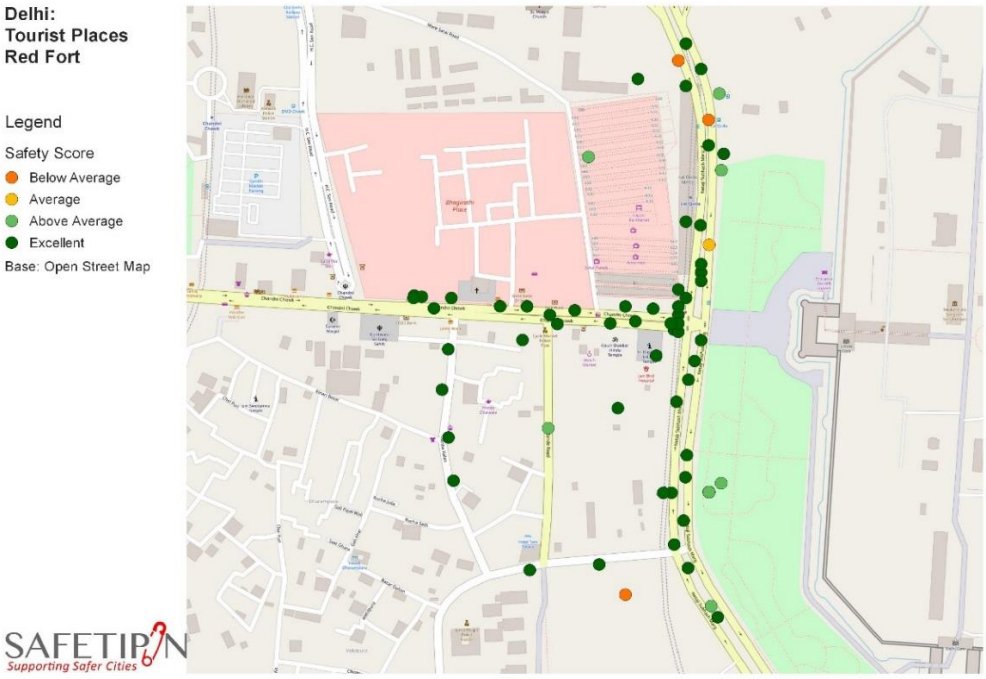
Figure 13: Safety Score – Red Fort



Source: Author

In the below map, points located around the tourist place as seen in the Safety Score map

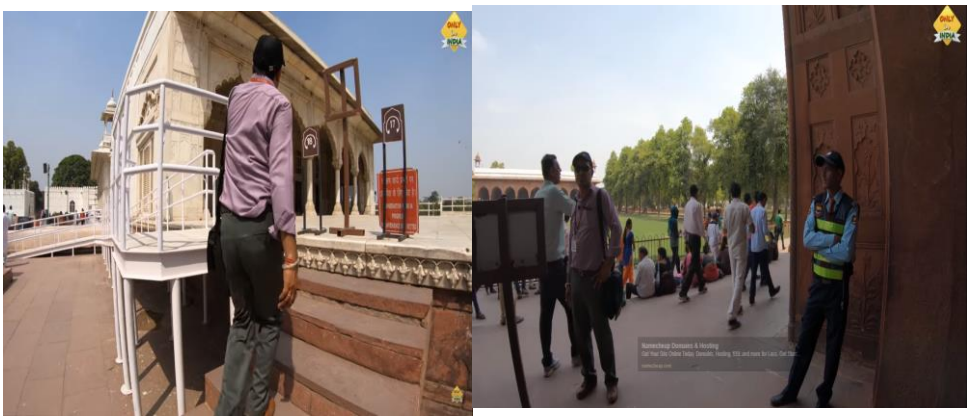
Map 5: Indicating Safety Score ratings around Red Fort



Source: Safetipin, 2020

Security is present at different locations inside the red fort and ramps are there to access in different parts and have good lighting, walkpath infrastructure.

Figure 14: Ramps and Handrails, private security and different locations inside red fort.



Source: Youtube, Virtual Audit.

Overall, this tourist place is highly secured, safe and accessible but still need to improve parameters related to accessibility.

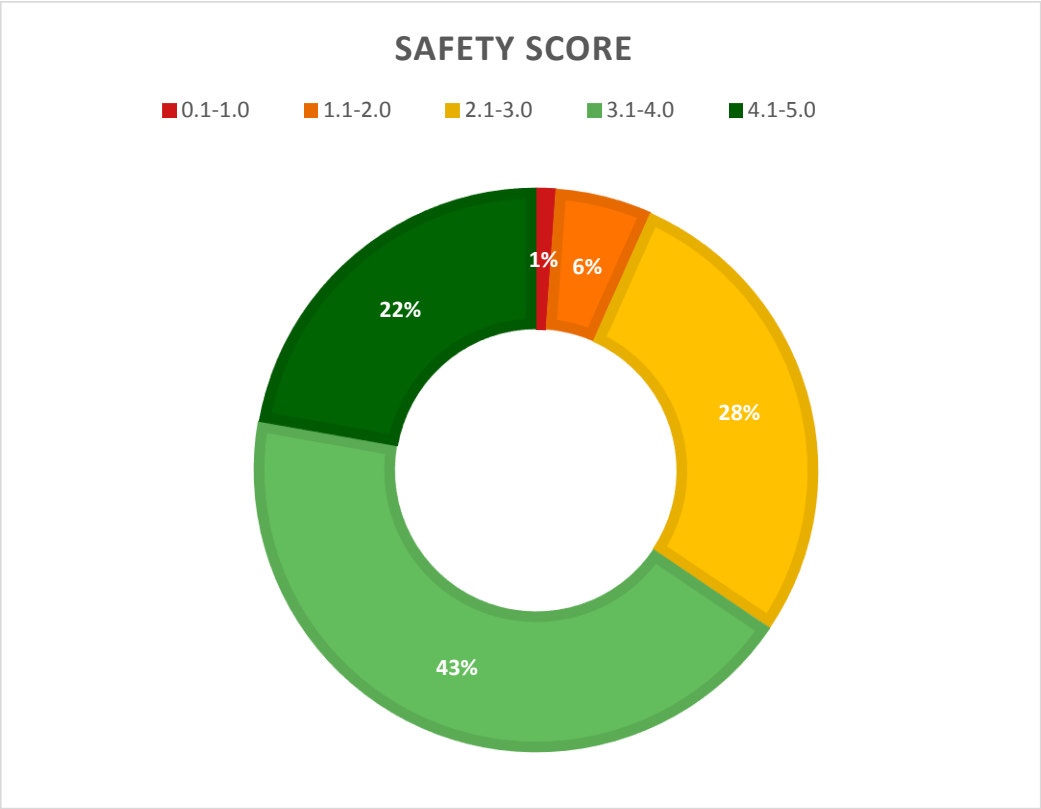
4.5 India Gate

The India Gate is located at the center of the New Delhi and the construction started in 1921 and completed in 1931 to remember British India soldiers who laid down their lives during first world war and third Anglo-Afghan war of 1919. India gate surroundings have lush green and well-maintained gardens. In these days it holds an importance not only as a tourist attraction but also a symbol of nationalism. Tourists mostly visits in the evening and night tourism because of fountains and colored lights also known as India war memorial.

Safety Score - Outside the tourist place

Overall safety score was rated as 3.5 which is above average and total 90 T- audit points.

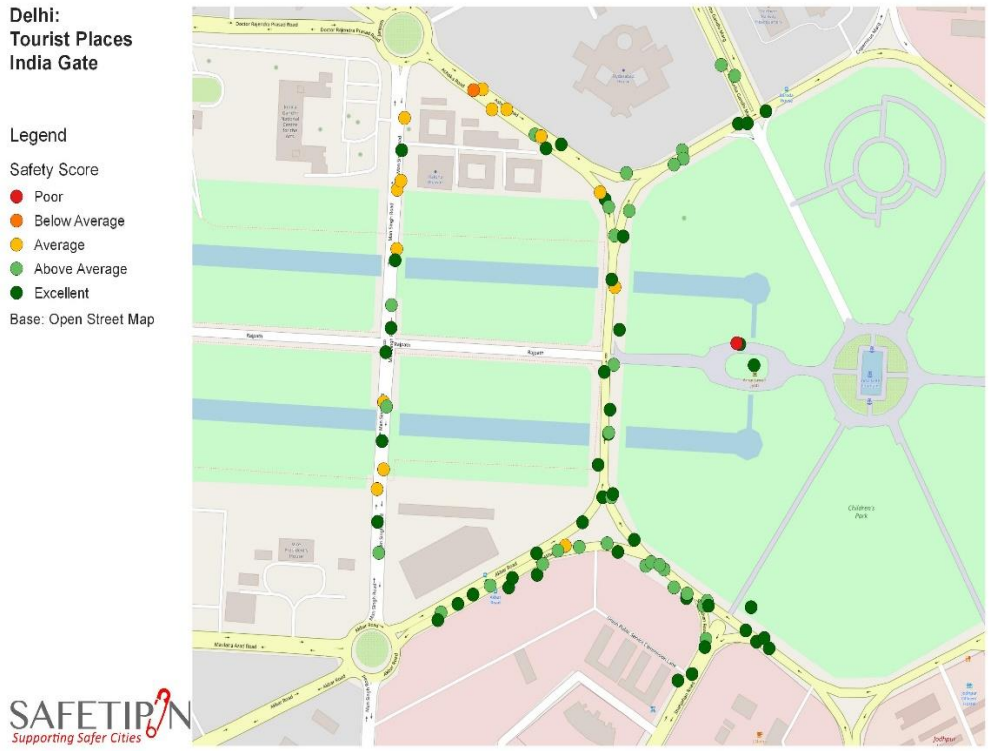
Figure 15: Safety Score – India Gate



Source: Author, 2020

In the below map, points located around the tourist place as seen in the Safety Score map

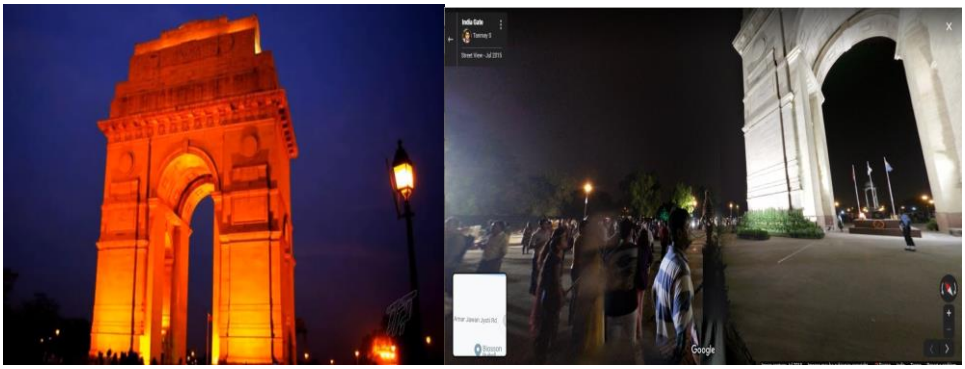
Map 6: Indicating Safety Score ratings around India Gate



Source: Safetipin, 2020

According to the results India gate is well equipped with walk path and lighting infrastructure even rental bikes are available to ride around the tourist place. It is extended to war memorial which has accessible facilities includes ramps, handrails and public toilets.

Figure 16: Night tourism at the India gate



Source: Google map street view.

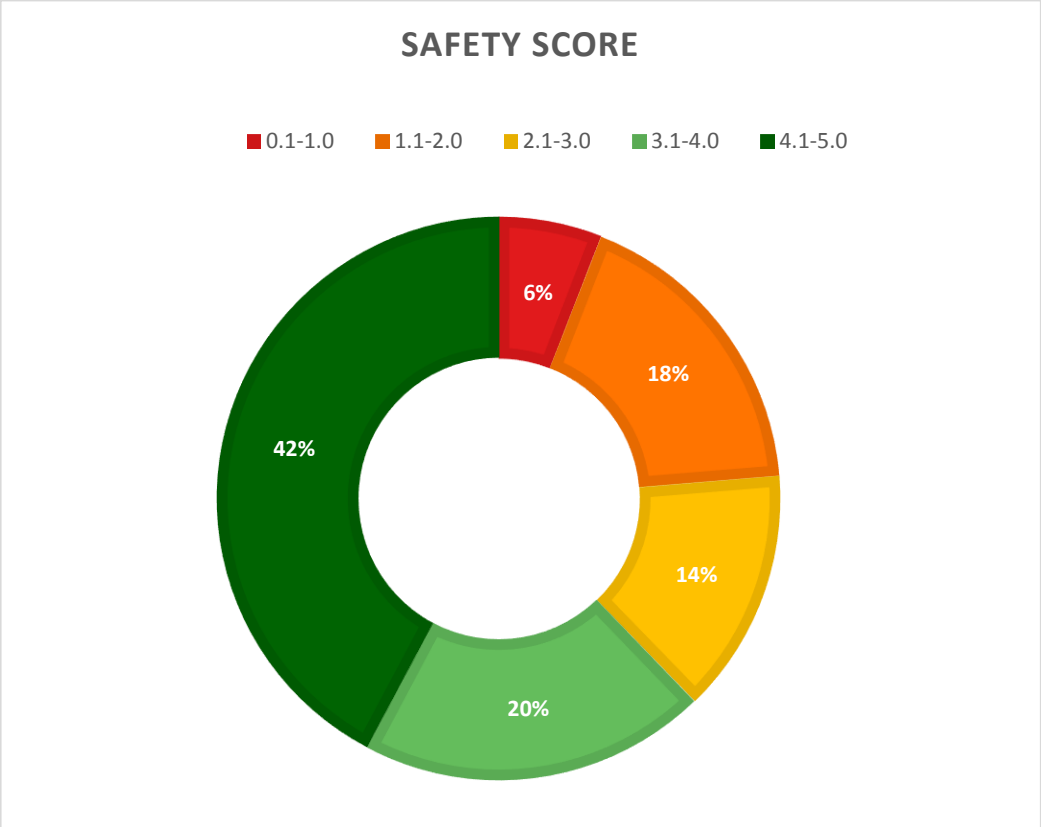
4.6 Humayun Tomb

Humayun tomb is located in Lodhi gardens and first garden tomb of Mughal emperor in Indian subcontinent. It was constructed in mid-16th century nine years after the death of Humayun. Inside the tomb is designed with garden squares with pathways water channels. Later in between 17th to the 19th century the garden was filled with the tombs of Humayun's descendants and his entourage and several Mughal emperors are even graved inside the walled tombs. Today this is emerged as the tourist destination and UNSECO identified this as the one of the world heritages sites in India.

Safety Score - Outside the tourist place

Overall safety score was rated as 3.3 which is above average and total 135 T- audit points.

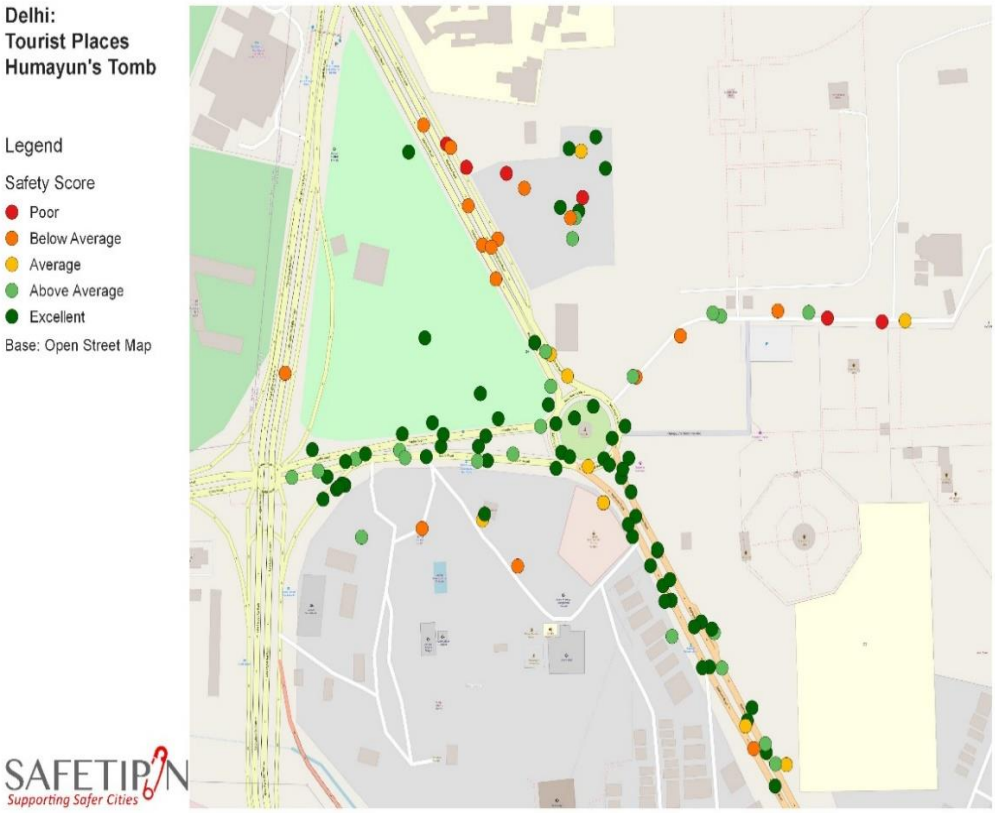
Figure 17: Safety Score – Humayun Tomb



Source: Author, 2020

In the below map, points located around the tourist place as seen in the Safety Score map

Map 7: Indicating Safety Score ratings around Humayun tomb



Source: Safetipin, 2020

Dedicated car parking is available at the tourist place. Tickets are available and at the entry gate which includes security check but didn't noticed any ramp or hand rails to access the tomb.

Figure 18: Car parking, ticket counter at the entry of the Humayun Tomb

Source: Youtube Virtual Audit.

Overall, this tourist destination is safe to visit but lack of accessible infrastructure makes hard to physically challenged tourist visitors.

4.7 Jantar Mantar

Jantar Mantar was built in 1724 by Maharaja Jai Singh of Jaipur who built this observatory went on to build other observatories in Jaipur, Mathura, Varanasi and Ujjain**.** The main use of the structure to gather astronomical tables which observe the time and movement of the sun, moon and other planets. It consists of different instruments which helps to predict time such as samara, Jayaprakash and Misra yantra. Now a days Jantar Mantar is one of the popular tourist destinations in the New Delhi.

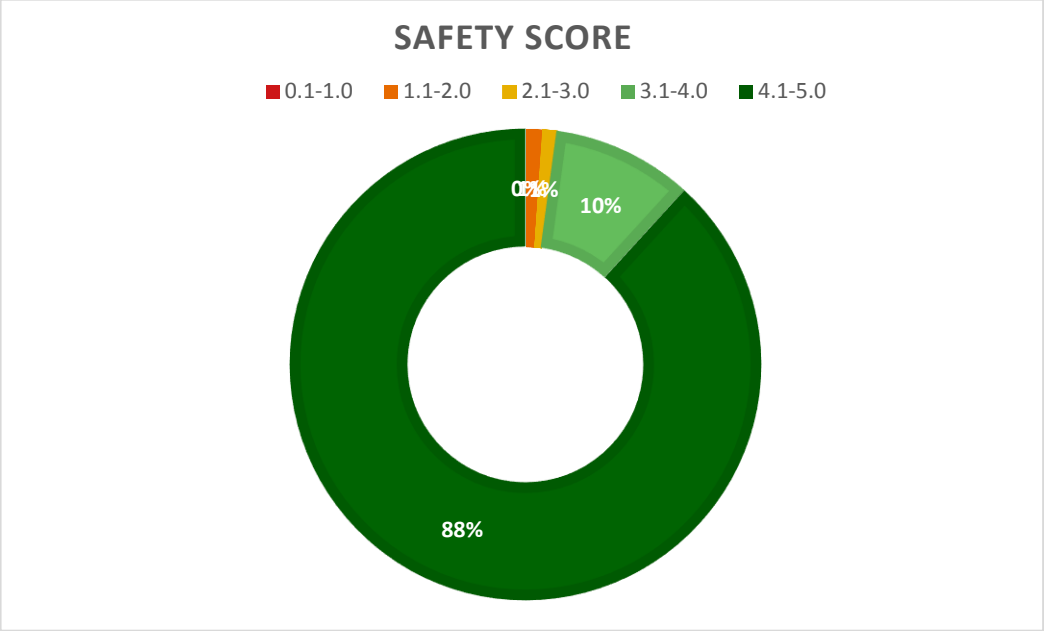
Figure 19: Walkpath inside the Jantar Mantar

Source: Google maps Street View.

Safety Score - Outside the tourist place

Overall safety score was rated as 4.3 which is above good and total 187 T- audit points.

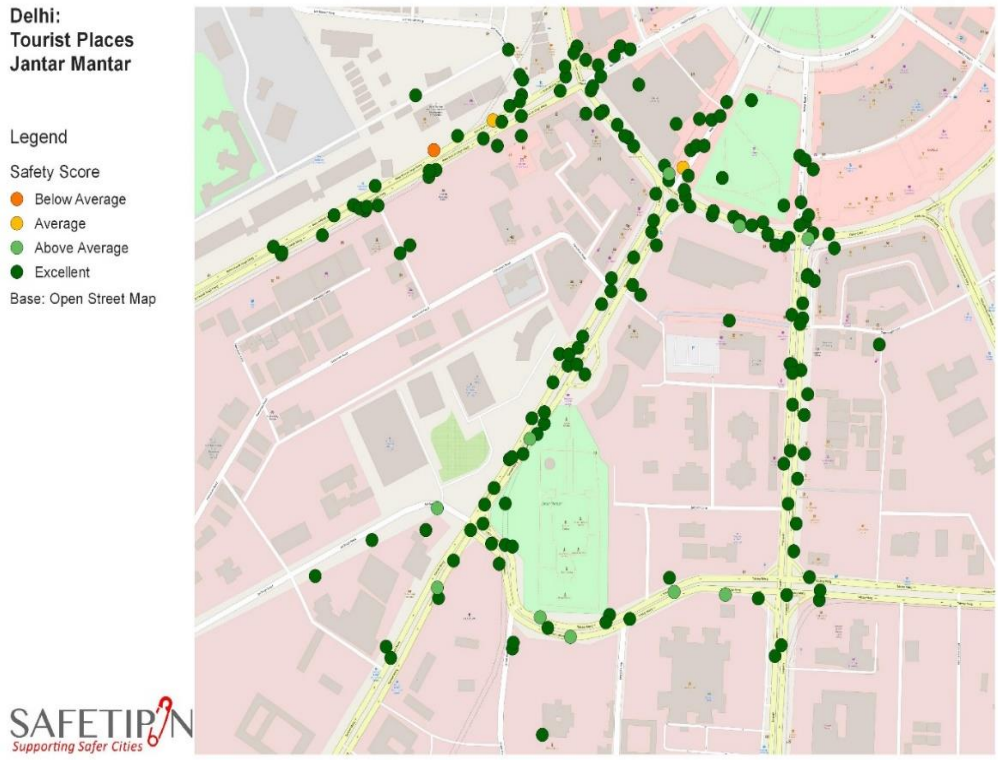
Figure 20: Safety Score – Jantar Mantar



Source: Author, 2020

In the below map, points located around the tourist place as seen in the Safety Score map

Map 8: Indicating Safety Score ratings around Jantar Mantar



Source: Safetipin, 2020

There is no dedicated car parking available and there is no hand rails or tactile pavement and private security is existing at the gate entry. Accessible to public toilets and drinking water available and wheel chairs available. Auto rickshaws near gate entry and street vendors are encroached the walkpath.

Figure 21: Street vendors and auto drivers encroached walkpath

Source: Google map Street View.

Overall, this tourist destination is safe to visit but lack of accessible infrastructure like tactile pavement makes difficult to physically challenged tourist visitors.

4.8 Purana Qila

Purana Qila also known as Old fort is one of the oldest forts in Delhi exists from pre- Mauryan period as Indraprastha and then later the present fort construction started in the time of Humayun at continued till Sher Shah Suri. In present days this fort is one of the tourist destinations in Delhi and famous for its night tourism sound and light show.

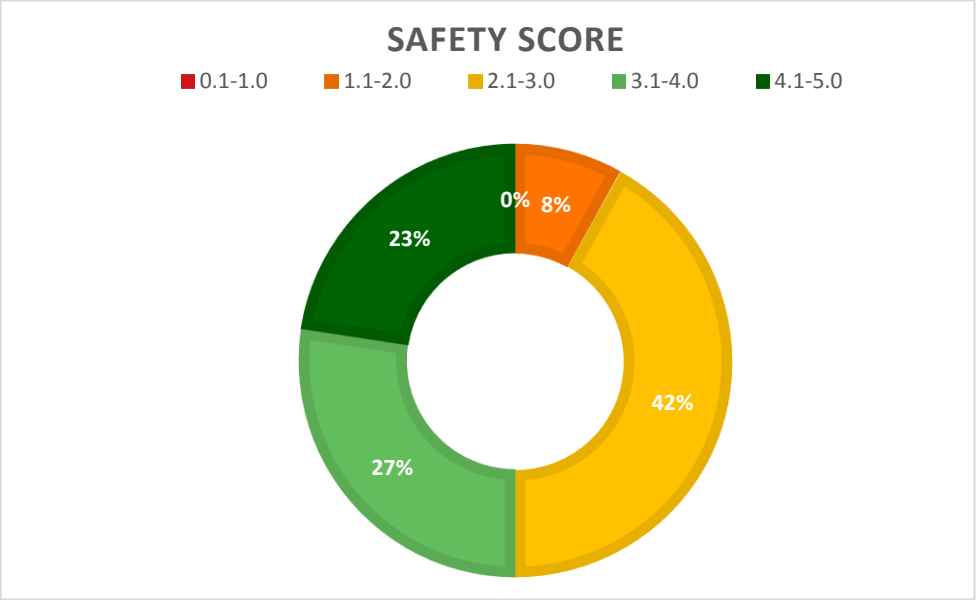
Figure 22: Ramps, handrails and tactile pavement inside.

Source: Google maps Street View.

Safety Score - Outside the tourist place

Overall safety score was rated as 3.3 which is above average and total 62 T- audit points.

Figure 23: Safety Score – Purana Qila



Source: Author

In the below map, points located around the tourist place as seen in the Safety Score map

Map 9: Indicating Safety Score ratings around Purana Qila

Source: Safetipin, 2020

Ramps and handrails are available to access different locations in the tourist place and some locations have tactile pavements. Women private security available at the entry and also patrolling done by security inside the tourist places as well. Overall, this tourist place is safe and accessible to visit.

Figure 24: Women security at the entry gate

Source: virtual tour and Street View.

4.9 Bangla Sahib Gurudwara

Bangla sahib gurudwara is one of the Sikh pilgrimage centers in Delhi located on Ashoka road near to Connaught place. In past it was the palace of raja jai singh was visited by Guru har Krishna sahib and cured people suffering with cholera and small pox by supplying water from a tank till exists today and contains medicinal properties in the 17th century. This pilgrim places also have prayer hall. Free food is served in the places where everyone shares a meal irrespective of religion.

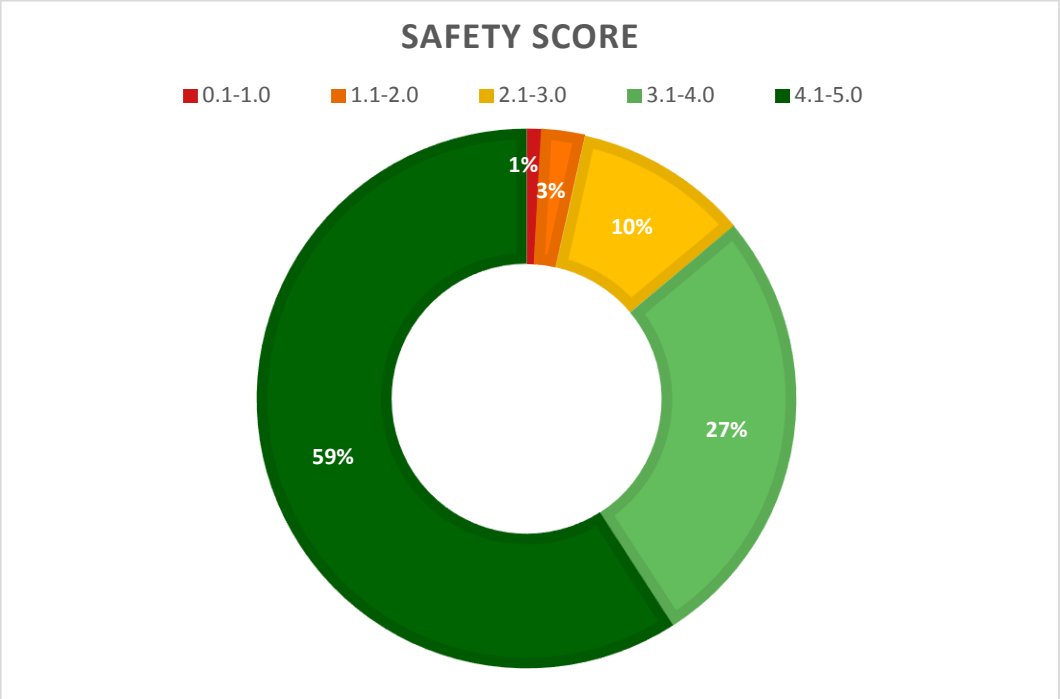
Figure 25: water tank and prayer hall inside – Bangla Sahib Gurudwara

Source: virtual tour and Street view

Safety Score - Outside the tourist place

Overall safety score was rated as 3.9 which is above average and total 115 T- audit points.

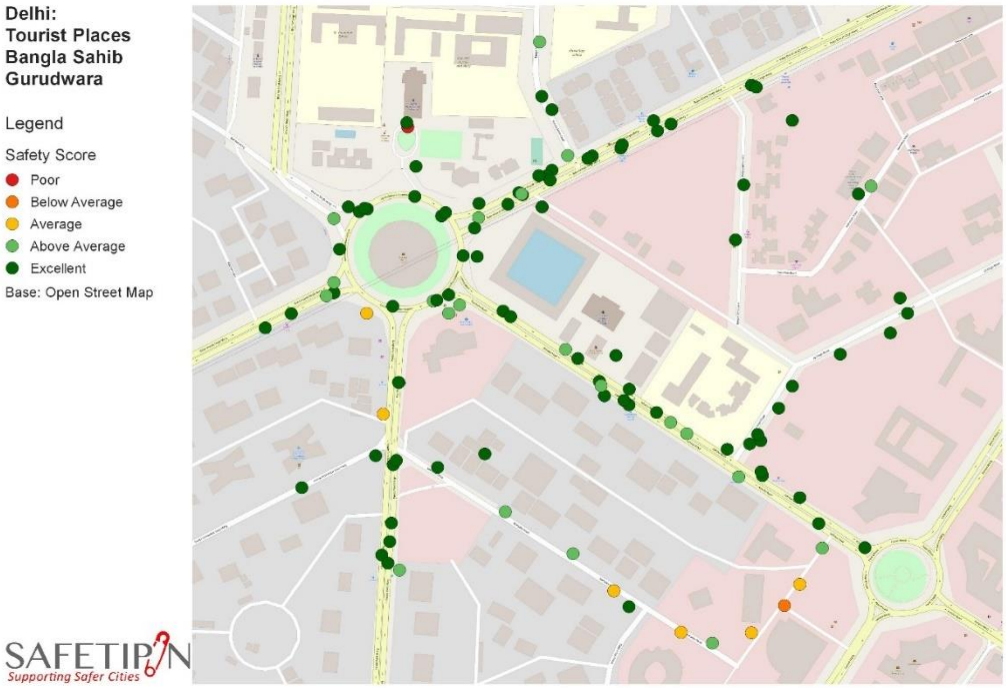
Figure 26: Safety Score - Bangla Sahib Gurudwara



Source: Author, 2020

In the below map, points located around the tourist place as seen in the Safety Score map

Map 10: Indicating Safety Score ratings around Bangla Sahib Gurudwara



Source: Safetipin, 2020

Ramps and handrails are available to access inside this place and Electric Cart is used as inside mobility for aged people. It is noticed that gender usage is high inside the Gurudwara. Public toilets available near the langar and drinking water is supplied to the visitors. The walkpath towards gurudwara was encroached by street vendors and informal auto-rickshaw stand.

Figure 27: Ramps, handrails and E -Cart inside the Bangla sahib Gurudwara.

Source: Virtual tour-Bangla Sahib Gurudwara

Overall, this place is safe in terms of safety and accessibility parameters but still need to improve the security at the entry exit gates.

4.10 Tughlakabad Fort

The Tughlakabad Fort was built by the founder of Tughlaq dynasty, Ghiyas-ud-din Tughlaq in early 14th century located near the Asola Bhatti Wildlife Sanctuary in Delhi. Today it is one of the historical forts in Delhi replicates the unique architectural works of Tughlaq dynasty.

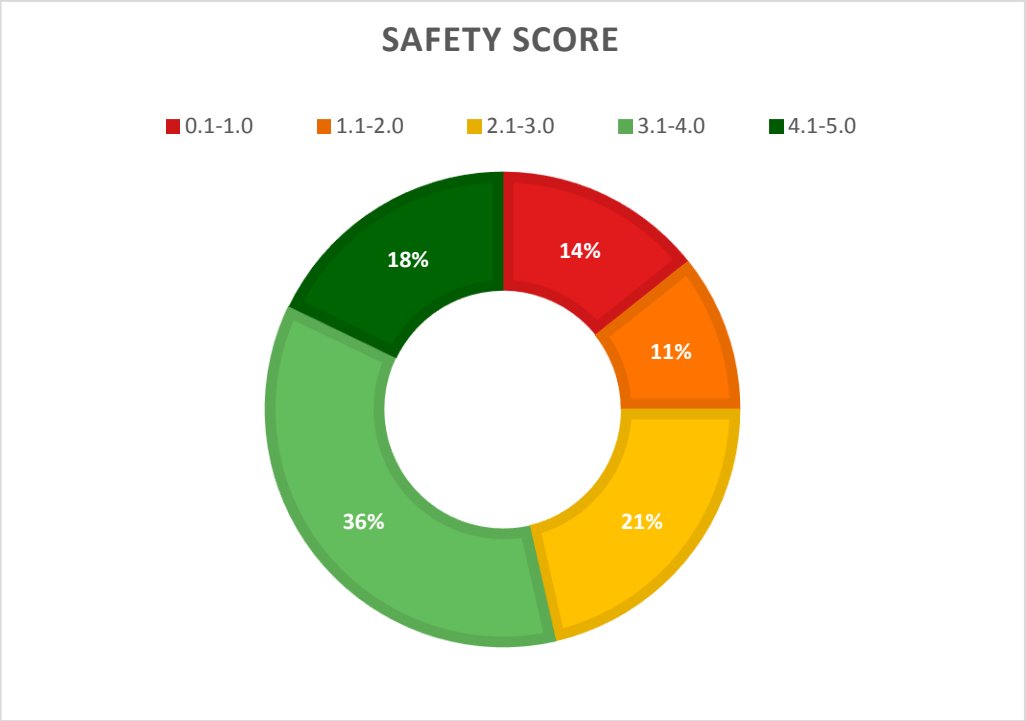
Figure 28: Inside View of the Tughlakabad fort

Source: YouTube, Virtual Tour.

Safety Score - Outside the tourist place

Overall safety score was rated as 3.0 which is average and total 28 T- audit points.

Figure 29: Safety Score - Tughlakabad Fort



Source: Author, 2020

In the below map, points located around the tourist place as seen in the Safety Score map

Map 11: Indicating Safety Score ratings around Tughlakabad Fort

Source: Safetipin, 2020

Ramps and handrails are not available to access inside this place even the walkpath is not good to access different location. It is very difficult to access this tourist place for the physically challenged people. Private security available at the entry and It is noticed that crowd and gender usage is very low inside the fort. Parking space is available at the fort entry.

Figure 30: Private security, ticket counter at the Gate entry and parking space.

Source: Youtube, Virtual tour

Overall, this tourist place is not safe in terms of safety and accessibility.

4.11 Key Findings:

In the below graph on basis of secondary data collected through manual audits and on basis of parameter averages and gaps are identified inside these tourist places.

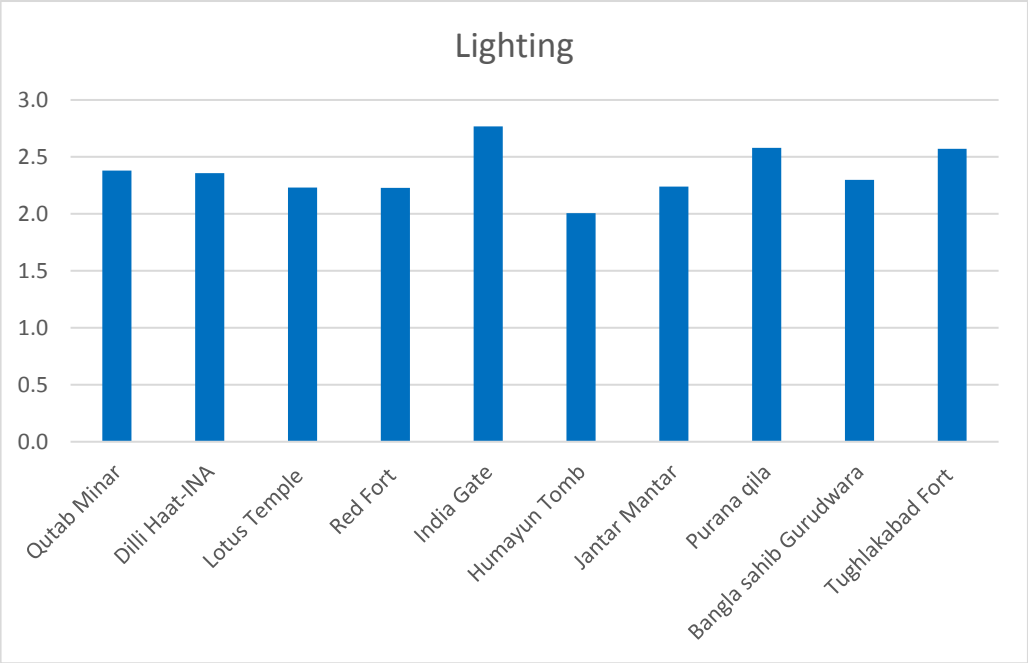
Table 4: Average rating of parameter by Safetipin.

1	0.1 – 0.6	Poor
2	0.7 – 1.2	Below Average
3	1.3 – 1.8	Average
4	1.9 – 2.4	Above Average
5	2.5 – 3.0	Good

Source: Safetipin

Lighting:

Figure 31: Lighting parameter average at around different tourist places.

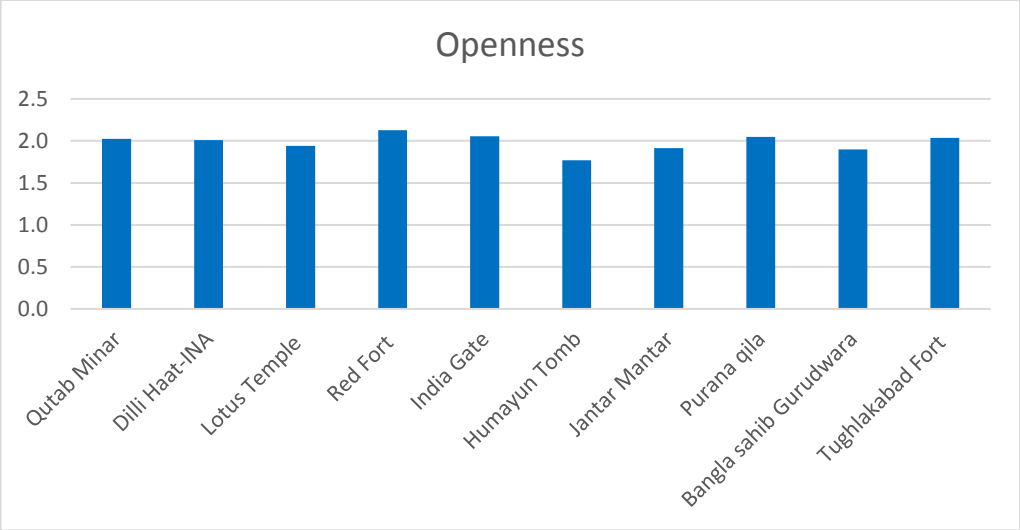


Source: Author

- All the tourist places are rated as above average.
- Lighting around India gate is rated as Good followed by Purana Qila and Humayun tomb is least rated.

Openness

Figure 32: Openness parameter average at around different tourist places.

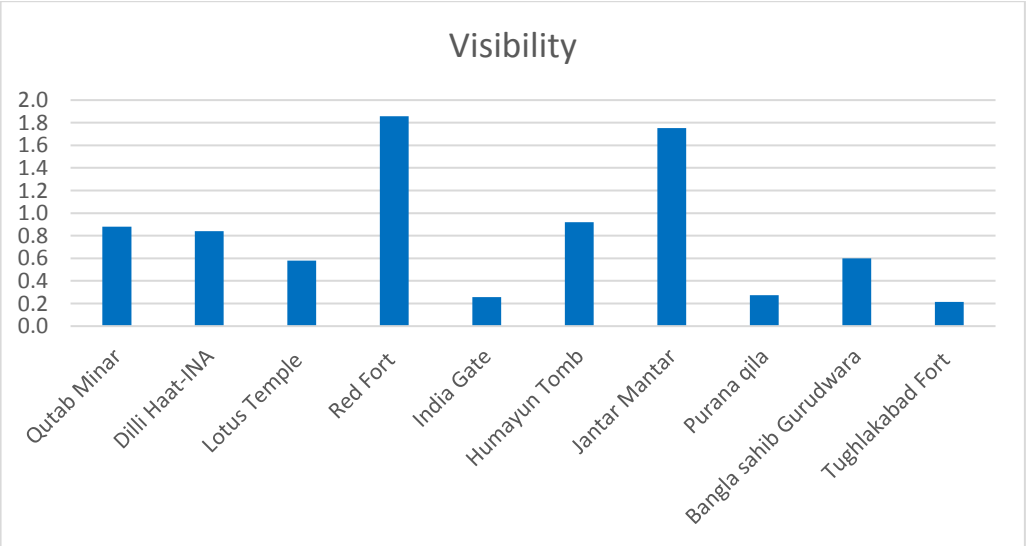


Source: Author

- All the tourist places are rated as above average and Humayun tomb rated as average.

Visibility

Figure 33: Visibility parameter average at around different tourist places.

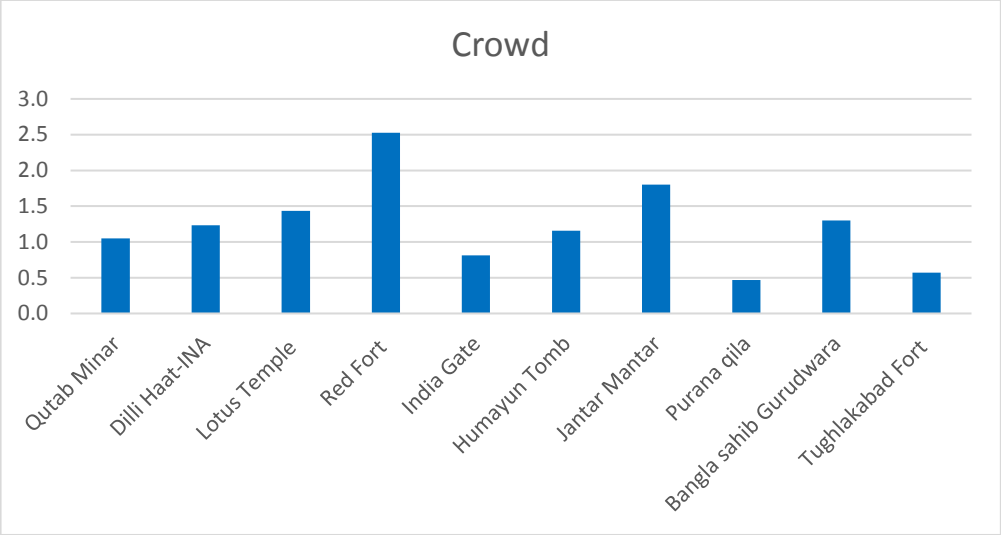


Source: Author

- Red fort and Jantar Mantar are rated as above average and average
- Tughlakabad fort and purana Qila rated as poor and India gate is not considered with visibility as it come special tourist protecting area.

Crowd

Figure 34: Crowd parameter average at around different tourist places.

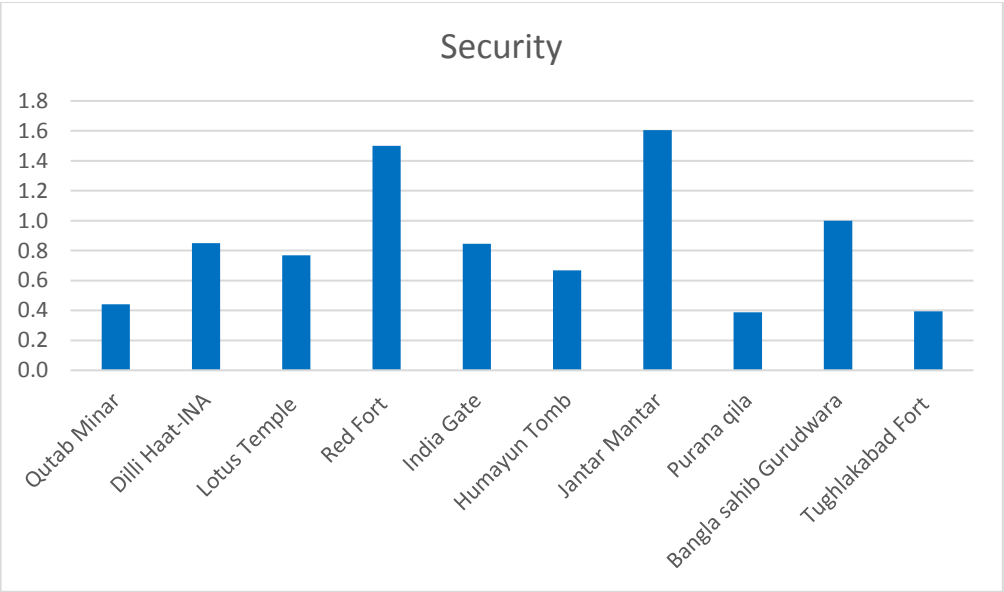


Source: Author

- Red fort is rated as good and Tughlakabad fort, Puruna Qila rated is poor.

Security

Figure 35: Security parameter average at around different tourist places.

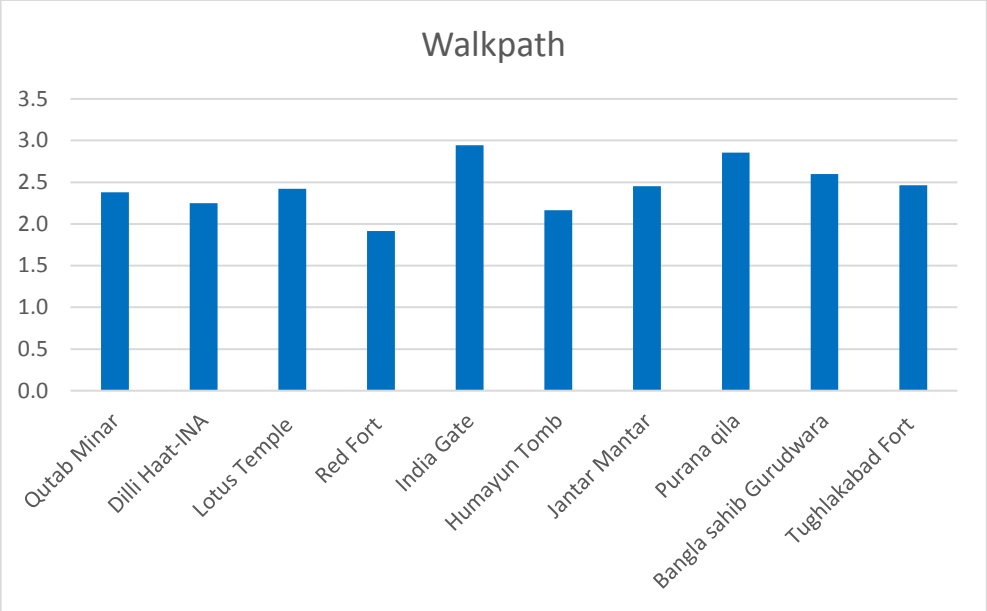


Source: Author

- Security parameter around Tughlakabad fort, Purana Qila, Qutub minar rated as poor. Red fort and Jantar mantar rated as average.

Walkpath

Figure 36: Walkpath parameter average at different tourist places.

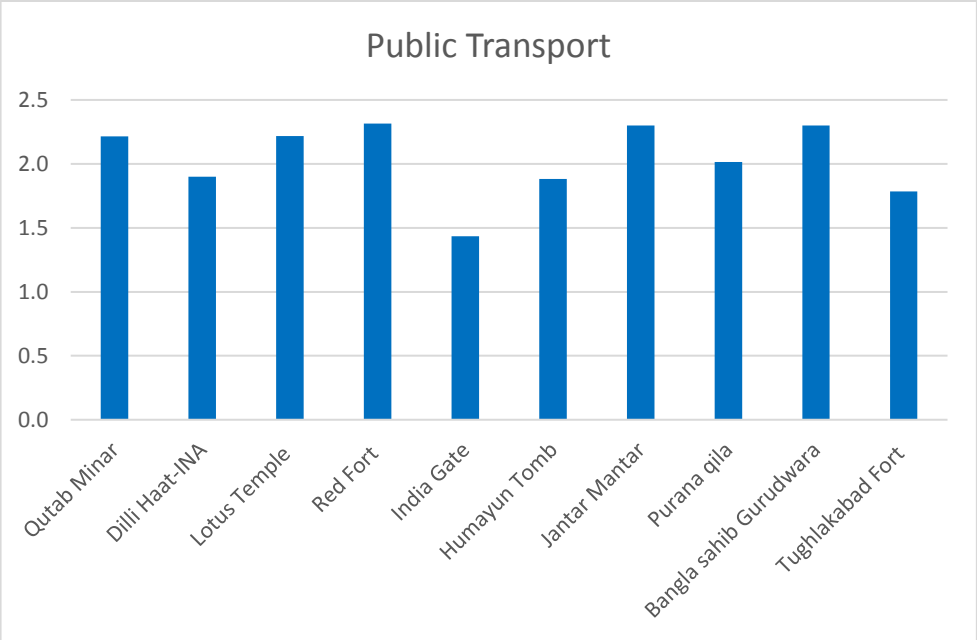


Source: Author

- Overall, around all tourist places are rated as Above average.

Public Transport

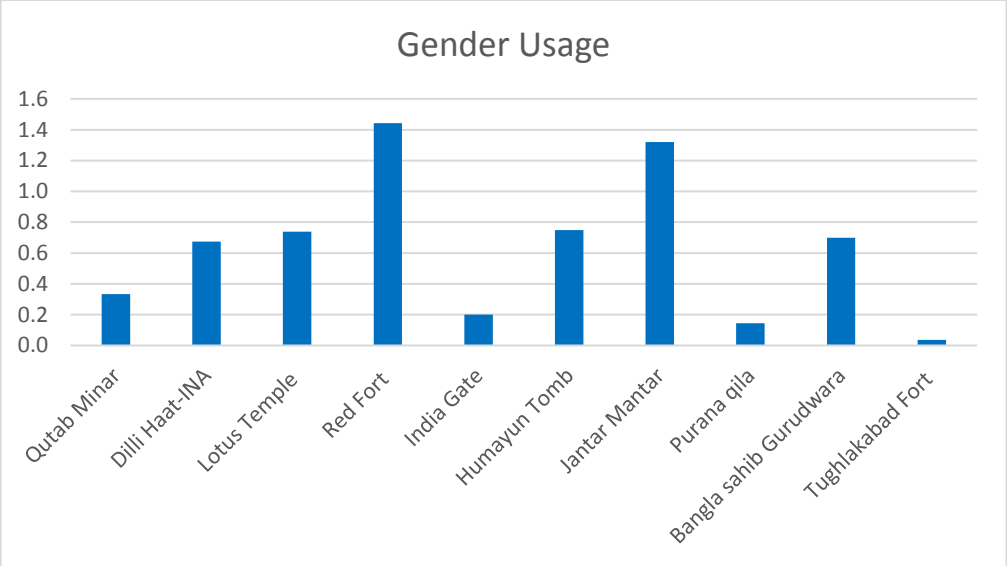
Figure 37: Public Transport parameter average at different tourist places.



Source: Author

Gender Usage

Figure 38: Gender usage parameter average at different tourist places.

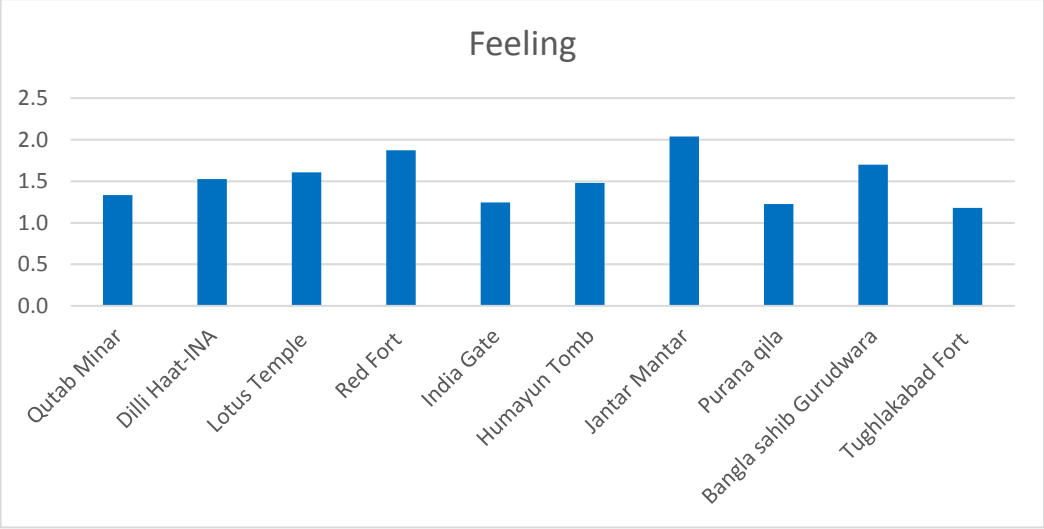


Source: Author

- If you observe the above figure gender usage is poor i.e. rated as 0 around the Tughlakabad fort.

Feeling

Figure 39: Feeling parameter average at different tourist places.



Source: Author

This is around the tourist places these parameters may vary at the entrance points because of the different activities and tourist policing patrols near this entry and exit points.

Table 5: Tourist place Findings -Safety and Accessibility

Tourist Place	Safety and Security	Accessibility
Qutab Minar	Tourist police and private police available at the entry points and continuous patrolling done inside.	Different locations even drinking water and public toilets are accessible. Wheel chairs and clock room available at the gate entry.
Dilli Haat-INA	Private security and police are available near the entry gate.	This market is designed with proper ramps and handrails here ever required like ticket counter at the public toilets.
Lotus Temple	Tourist post outside the and police, private security available at the gate entry and also have continuous surveillance inside.	Ramps and handrails are provided to access inside the temple building and paved walkpath are available to access different locations.
Red Fort	Specially protected by armed forces as well as Delhi tourist police. Private security patrolling inside.	Some monument and buildings are installed with the ramps and handrails. Paved walkpath to access different locations.
India Gate	Specially protected by armed forces as well as Delhi tourist police.	No accessible issues identified.
Humayun Tomb	Security check at the entry gate and tourist post outside. Private security patrolling inside.	Still some locations like monuments are not accessible and didn't find any ramps and handrails to access.
Jantar Mantar	Not identified tourist police and no proper checking at the gate entry.	Due to its location advantage no need of accessible infrastructure.
Purana Qila	Police security outside the tourist place and patrolling	No accessible issues identified.

	by private security at gate entry and inside	
Bangla sahib Gurudwara	Private security and police are available near the entry gate.	Ramps and handrails and E-Cart are provided to access inside the temple building and paved walkpath are available to access different locations.
Tughlakabad Fort	No security noticed inside the tourist location.	Not identified any ramps, handrails or even proper walkpath to access the different locations.

Source: Author

General Recommendations:

1. Improve Lighting infrastructure around the tourist places like at parking spaces, public toilets, pedestrian walkpath leading to public transport and para transit facilities and inside the tourist place in different location. If required add more street light infrastructure installed to make the surroundings around the tourist place to brightly lit night.
2. Individual tourist places should be audited with physically disabled persons and identify the accessibility infrastructure requirement on demand basis and wheel chairs provided at the tourist place
3. Install the ramps, handrail and tactile pavements where every necessary to access the public infrastructure at ticket counters, security check points, drinking water facilities and at public toilets inside the tourist places according to standard guidelines.
4. Dedicated street vendors zones because in most of the tourist places walkpath encroached by the street vendors block and restricts the movement of tourists. So, place making and planning is important at the entry and exit gates and promote the night tourism to make sure activity around the places that may increase the feeling of safety.
5. Para transit facilities must be provided near the tourist places. In our study it is observed that auto rickshaws are parked informally on the roads which restricts the flow of traffic and movement of tourists. Dedicated auto stands

are planned near the parking space or at bus stops with in a walkable distance.

6. Walkpath infrastructure should be improved from primary public transport facility if possible, from the metro stations with continuous tactile pavement path leading to tourist place.
7. Increase the Tourist police patrolling and private patrolling around the tourist place, if possible, select and train local volunteers as a local response team.
8. Install advanced AI- surveillance mechanism like pattern, face recognitions and Vehicle number plate recognition in cameras and online/ offline feedback mechanism to improves the quality of experience at the tourist places.
9. Develop standardized indices to measure and compare safety and accessibility across tourist places at a national level. This can be used to rank tourist laces as well as analyze the gaps present in a much more systematic and detailed manner to improve overall tour experience quality.

Table 6: Recommendations – Site Specific

Tourist Place	Recommendations
Qutab Minar	• Focus on planning place making outside the tourist place for food and other activity purpose. • Develop the feed back system to capture the tourist visitors experience. • Temporary tactile flooring to make this location accessible for the blind visitors.
Dilli Haat-INA	• As this is multipurpose market place install continuous tactile pavement and also provide wheel chairs.
Lotus Temple	• Focus on planning place making outside the tourist place for food and other activity purpose and also install tactile pavements inside the tourist place.
Red Fort	• Install temporary tactile flooring to make this location accessible for the blind visitors.
India Gate	• Install accessible infrastructure on walkpath.

Humayun Tomb	• Install tactile pavements, ramps and handrails to access the monuments and increase the security patrolling inside the tourist place.
Jantar Mantar	• Provide dedicated parking spaces near the location. • Improve security and install temporary tactile pavement inside.
Purana Qila	• Focus on planning place making outside the tourist place for food and other activity purpose to support the night tourism.
Bangla sahib Gurudwara	• As it is religious place need to increase the security in different locations.
Tughlakabad Fort	• Install tactile pavements, ramps and handrails to access the different locations inside the fort and increase the security patrolling inside the tourist place.

Source: Author

Limitations:

1. All the study done on the basis of secondary data available with safetipin organization and internet (Virtual tours) due to Covid19 pandemic situation.
2. Application limitation like it is used to audit the public spaces so to take the feedback from the tourist need to explore new methods of data collection.
3. Full data not available related to infrastructure facilities, security and surveillance mechanisms due to lack of interaction with primary stake holders at the selected tourist place.

Conclusion:

Developing countries like India with unique historical place, beautiful forests, mountains, coastal area and mega cities has a great potential to emerge as one of the world's premier tourist destination in the future which contributes to the country GDP. Ministry of tourism and Archeological survey of India conducting research and development in different sectors related to tourism and improving the institutional and infrastructural capacities to provide good experience to the domestic and international tourist visitors. Initiatives like tourist policing implemented and commenced at every tourist place and made information accessible online still lack of infrastructure and leaving informal sectors unorganized creating kiosk and rising issues related to safety and accessibility around the tourist places. It's the time to upgrade with technological and planning interventions at the tourist destinations. To develop any kind of infrastructure identifying demand is difficult and for that data collection tools and methodologies are required to analyze the actual demand. Organizations like safetipin is working on developing these standard data collection tools to enhance the safety of a citizen. By promoting night tourism and improving the infrastructure facilities in Metropolitan cities like Delhi will improve the quality of life for the localities dependent and residing near the tourist places

Scope of future research:

Safety, Security and accessibility is the main concern for every tourist individual at the tourist destinations. A national level Tourist place centric data collection tools should be developed for different type of tourist places to access the infrastructure requirements. This study is based on some literature review and guidelines and it can be improved through research.

Major Project Learnings

In this major project I learned the historical, cultural, heritage and economic importance of the tourist places and Social usage. Importance of safety and accessibility at the tourist places and impact on the quality of experience due lack of infrastructure. Government initiatives to improve the security and surveillance by enforcing policy like tourist policing and police post at the tourist places.

Learned data collection methods using google forms and auditing public spaces physically and virtually using safetipin applications like MySafetiPin and Safetipin Site. Qualitative and quantitative analysis of data and developing questionnaires based on the requirement of the studies.

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Annexure

Table 1: Questionnaires to audit inside the tourist places

Sl.No	**Questions**
1	Video
2	Virtual tour (Y/N)
3	Availability of security or staff inside the tourist place?
4	Patrolling by security and staff?
5	Comment if any other issues are identified related to safety and security ?
6	Any opening point apart from the designated gates inside the tourist place?
7	How many opening points are there inside the tourist place?
8	Is there any security staff or surveillance present at those particular openings?
9	Upload the images of the openings
10	Comment if any other issues are identified at those openings?
	Accessibility
1	Which of the following are available inside the tourist place?
	Tactile flooring on walk-path (temporary or tactile mat in historical places)
	Ramps to access different locations in tourist place
	Handrails
	Others
2	Upload reference Images
3	Comment if any other issues are identified related to accessibility?
	Public toilet
1	Type of toilets available inside the tourist place?
	Mobile toilets
	Public toilets
	None
2	Which of the following facilities are present for the disabled?
	Separate toilet for disabled
	Ramp
	Handrail
	Tactile pavement
	Signages
	Other
3	Upload the images of the public toilets
4	Comment if any other issues are identified at the public toilets ?
5	Presence of the drinking water facilities ?(Y/N)
6	Upload the images
	Lighting
1	Rate the lighting inside the tourist place?(1/2/3/4)
2	Source of lighting inside the tourist place
	Light posts
	Advertisement/ Information light boxes
	Other
3	Upload images of the lightings
4	Comment if any other issues are identified related to the lighting infrastructur
	Walk-path
1	Rate the Walkpath inside the tourist place?(1/2/3/4)
2	Upload images of the walkpath
3	Comment if any other issues are identified related to Walkpath ?
	People and Gender usage
1	Number of people inside the tourist place ?(1/2/3/4)
2	Presence of women and children inside the tourist place ?
3	Upload the images
4	Comment if any other issues are identified related to tourist place ?

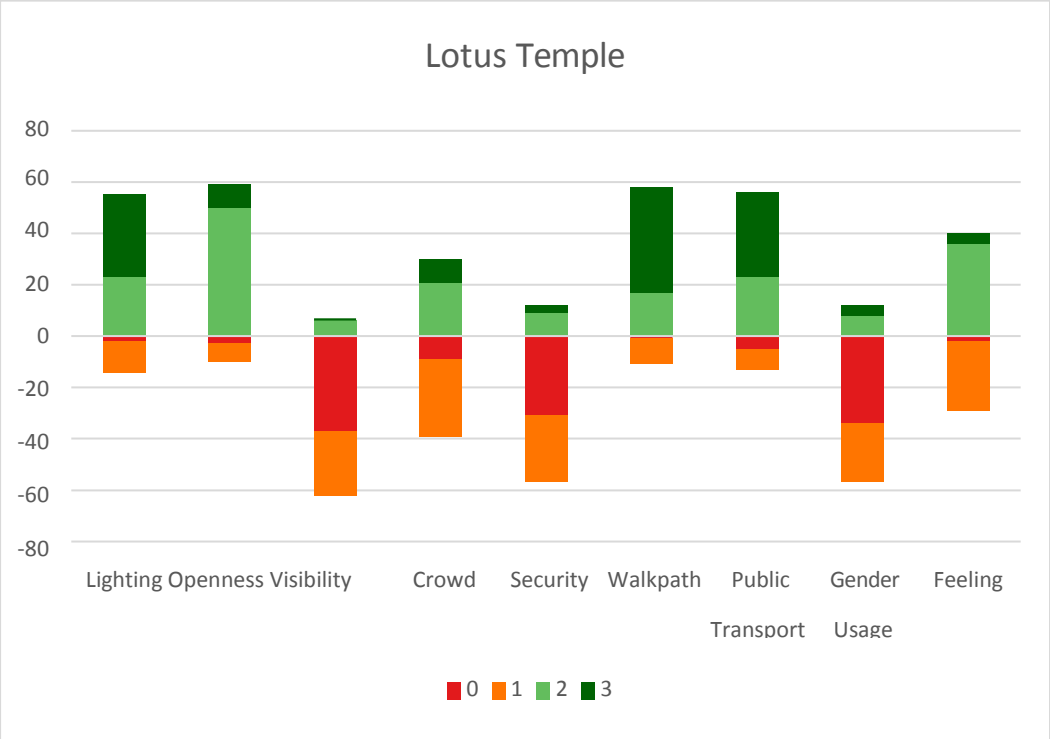
Table 2: Questionnaires to audit outside the tourist places

Sl.No	**Questions**
1	Video date
2	Virtual Tour (Y/N)
3	Name/ Number of the gate? (Ex: Entrygate01 /Exitgate01)
4	Are the entry and exit gates the same? (Y/N)
5	Which of the following are present at the gate?
	Permanent shops
	Temporary vendors
	Women Vendors
	Others
6	Closing timings of the permanent Shops ?
7	Upload the images permanent shops/temporary shops
8	Comment if any other issues are identified ?
	Security
1	Which of the following are available outside the tourist place?
	Police Check Post/ Barricade
	Private Security
	CCTV
	None
	Others
2	Upload the images
3	Comment if there are any other issues identified related to security?
	Transport
1	Which type of transport is available within 500mts distance from this gate?
	Metro Station
	Bus Stop
	IPT Stand (Auto/E-Rickshaw)
	Rental Bicycles Stand
	Other
2	Which of the following IPT are available ?
	Auto-Rickshaw
	E-Rickshaw
	Cycle -Rickshaw
	Van
	Other
3	Any particular space dedicated to the IPT? (Y/N)
4	Upload the images of the IPT
5	Comment if there are any other issues identified related to IPT?
	Parking Space
1	Which of the following are available?
	2 - wheeler
	4 - wheeler
	Bicycle
	Reserved Parking for the Disabled
	Other
2	Which of the following are available at the parking space?
	Lights at parking area
	Security check
	CCTV surveillance
	Parking Staff inside
	Parking staff at the entry
	Others
	Comment if any other issues are identified in the parking space?
	Upload the images of the parking space
	Did you notice any informal parking at the tourist place?
	Upload the images of the informal parking

Table 3: Questionnaires to audit outside the tourist places

	At the Gate
1	Is there the gate accessible within 50mts distance from the ticket counter? (Y/N)
2	Which of the following are available at the ticket counter?
	Availability of QR Scan code for online ticket booking
	Separate ticket counter for disabled
	Signages near the ticket counter
	other
3	Upload the images of the ticket counter
4	Condition of the path from the ticket counter to the gate?
5	Upload the images of the path from the ticket counter to the gate
6	Which of the following are available to access the gate?
	Ramps
	Handrails
	Tactile pavements
	Dedicated gates for disabled
	Other
7	Which of the following are available at the gate?
	Police/ CRPF /Women police
	Private secuirty and staff
	CCTV
	Other
8	Comment if any other issues are identified at the gate?
9	Upload images of the gate

Figure: Parameter-wise Pin Distribution – Lotus Temple

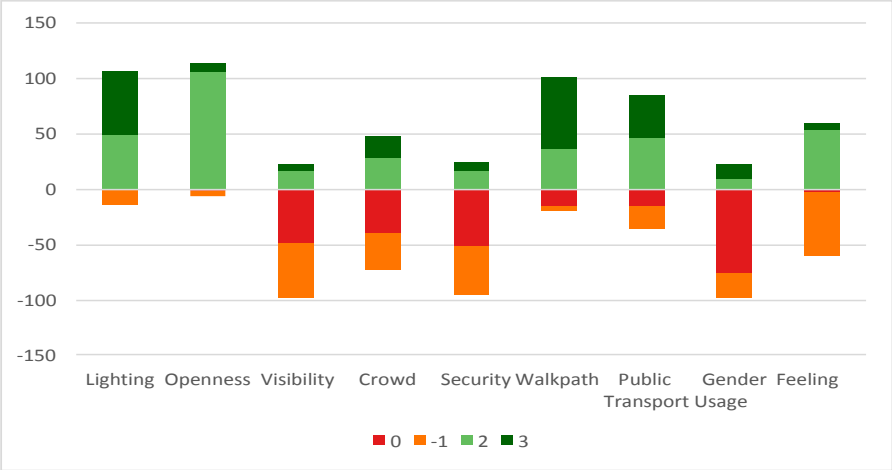


Source: Author

Figure: Parameter-wise pin Distribution – Qutub minar

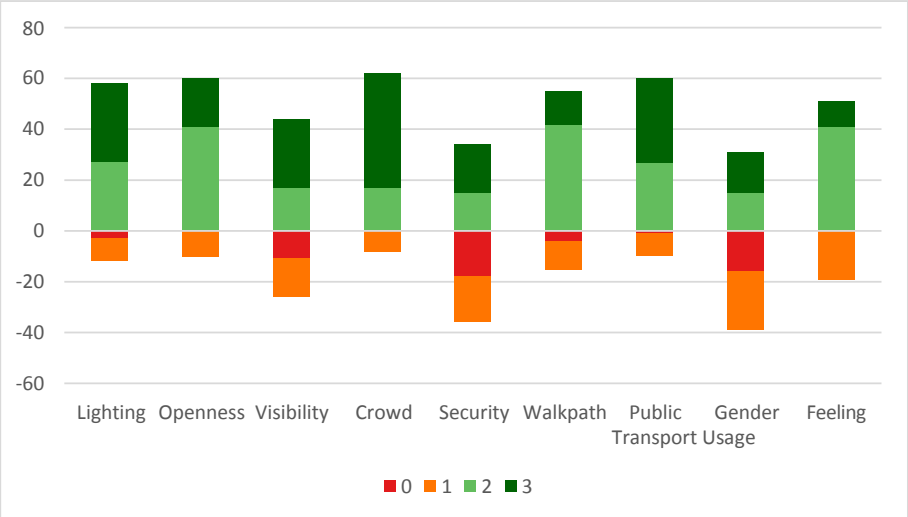
Source: Author, 2020.

Figure: Parameter-wise Pin Distribution – Dilli Haat INA



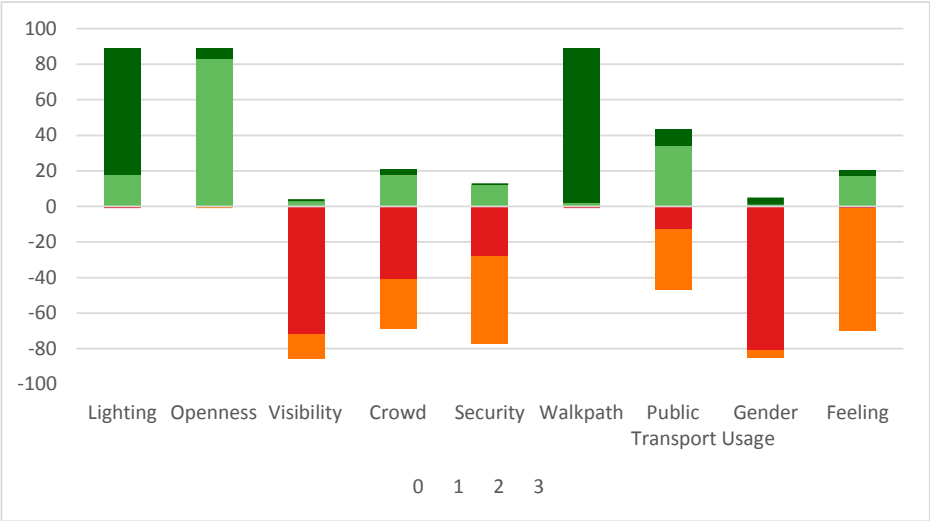
Source: Author, 2020

Figure: Parameter-wise Pin Distribution – Red Fort



Source: Author, 2020

Figure: Parameter-wise Pin Distribution – India Gate



Source: Author, 2020

Figure: Parameter-wise Pin Distribution – Humayun Tomb

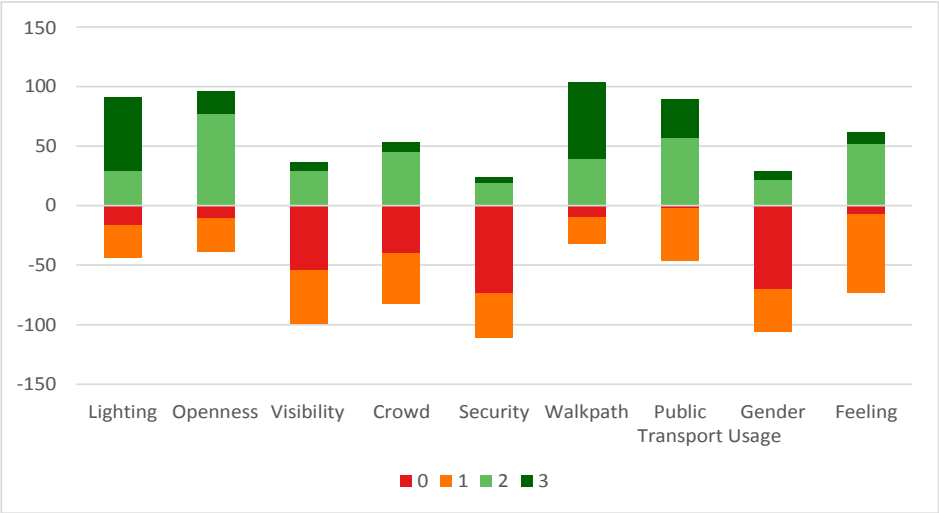


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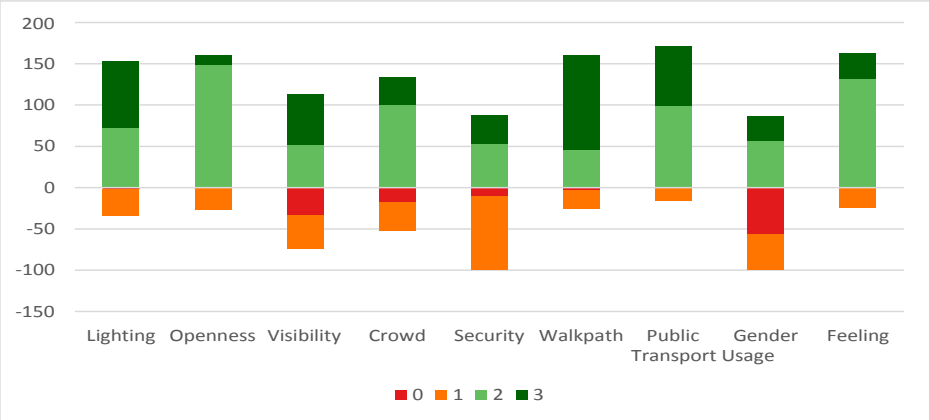
Figure: Parameter-wise Pin Distribution – Jantar Mantar



Source: Author, 2020

Figure: Parameter-wise Pin Distribution – Purana Qila

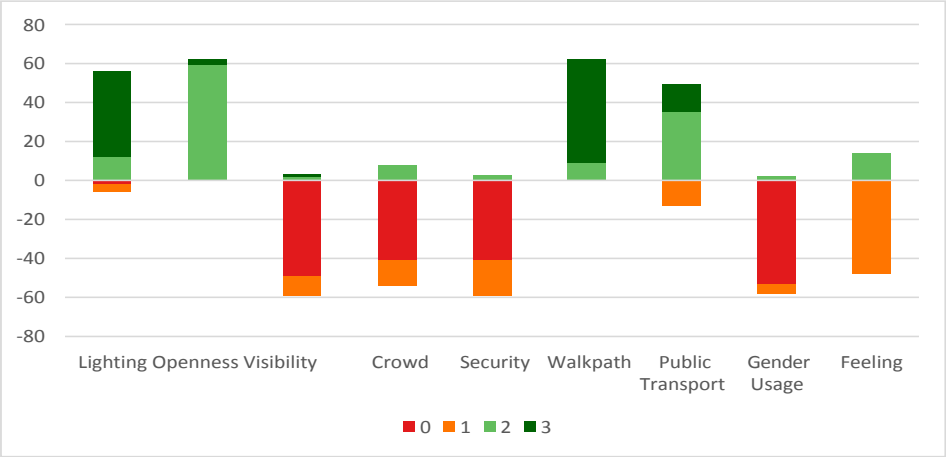


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Figure: Parameter-wise Pin Distribution – Bangla Sahib Gurudwara

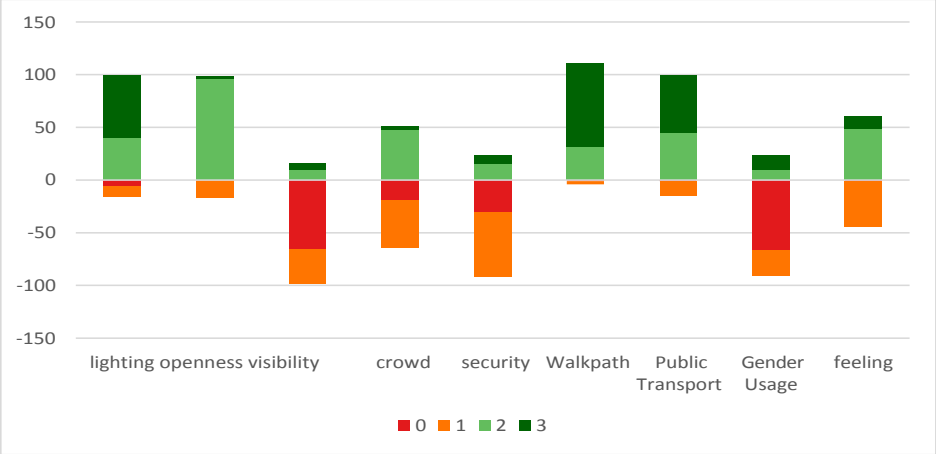

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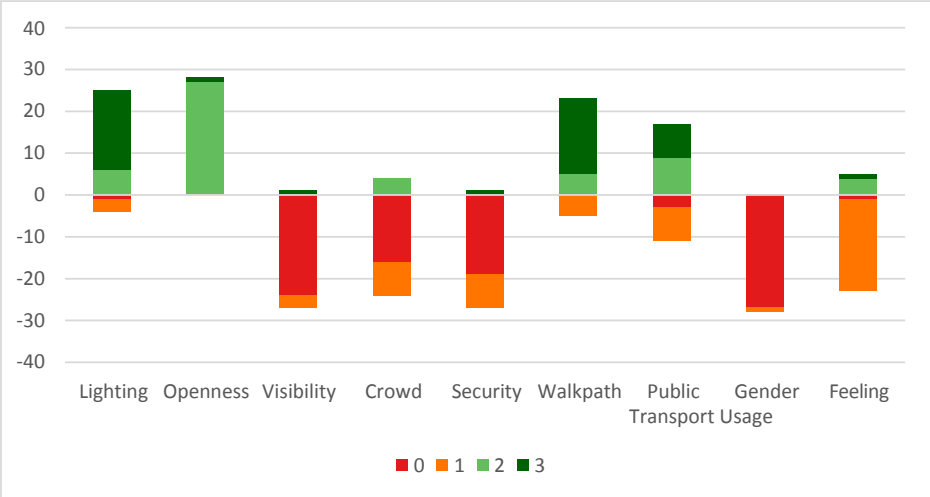
Figure: Parameter-wise Pin Distribution – Tughlakabad Fort



Source: Author, 2020